TM Forum Specification

Software and Compute Entity Management API User Guide

TMF730

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Introduction

The following document is the user guide of the REST API for Software and Compute entity management. It includes the model definition as well as all available operations.

Overview

The Resource domain covers *Physical* and *Logical* managed entities. Increasingly physical network appliances are being replaced by virtualized counterparts realized as software hosted on general purpose computing infrastructure. It is important that these software entities form part of a management continuum where physical and logical resources are equally well enabled.

This API introduces the data schemas required to manage software and compute entities using the Resource Domain Open APIs.

Concepts

The basic building block concepts needed for managing software and compute entities are:

- Software
- Function
- Connection Point
- Process
- Host

Software is a computer program, a sequence of instructions in a programming language. A ***Function*** is a capability of a software which transforms input(s) into output(s). A ***Connection Point*** makes input and/or output for a *Function* available and may be realized by an application program interface (***API***). A ***Process*** is an instance of execution of *Software*. A compute environment for software *Processes* is a ***Host***.

Software provides *Functions* which provide Connection Points. Any complex system may be decomposed into an interconnected structure of these basic elements.

Data Model

The above concepts are realized with the data model depicted in Figure 1 below. The *Entity/EntitySpecification* pattern, common to the TM Forum Open API data model, is applied for each entity type.

An instance of ***InstalledSoftware*** represents one or more *Processes* which are either currently executing, or may potentially be executed in the future, and is described by a ***SoftwareSpecification***. ***ResourceFunction*** instances are provided by *InstalledSoftware* and described by ***ResourceFunctionSpecification***. The *Connection Point(s)* of a *ResourceFunction* are realized by ***API*** instances which are described by ***APISpecification***.

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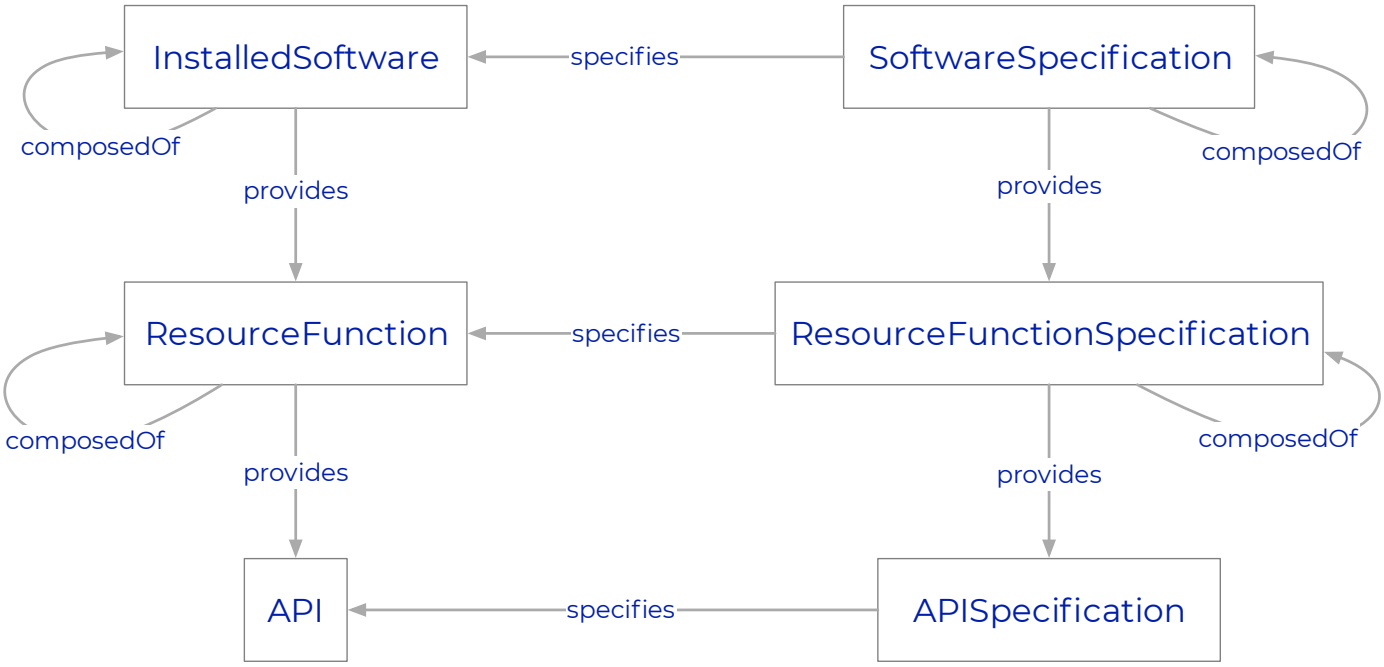


Figure 1: Software and Compute Entity Relationships

A *SoftwareSpecification* may reference an external ***SoftwareSupportPackage***, which provides the files containing the programs required for instantiated *Processes* of the *InstalledSoftware*, and which may be described by a ***SoftwareSupportPackageSpecification***. A ***HostingPlatformRequirement*** provides the requirements on a *Host* which must be fulfilled to run the *Software* and is described by a ***HostingPlatformRequirementSpecification***.

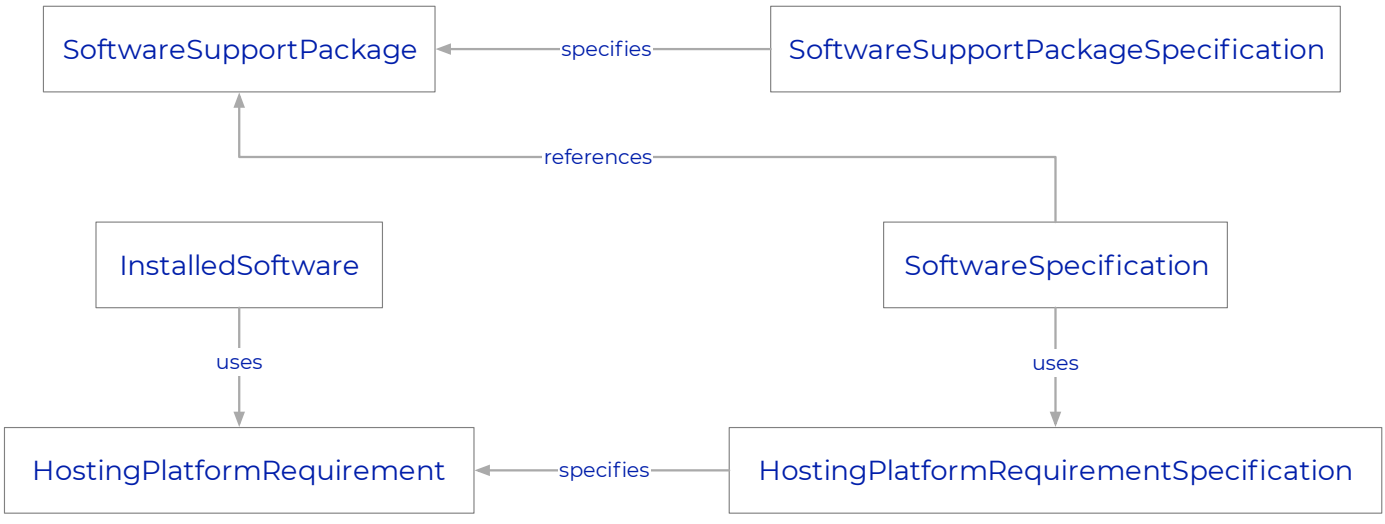


Figure 2: Additional Entities and Relationships

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SAMPLE USE CASES

5G Core Network Functions

A 5G Core network slice subnet contains *Functions* such as the *Access and Mobility Management Function (AMF)*.

The following contrived example depicts one possible model for an AMF implemented as a Kubernetes cloud native function (CNF). Here the CNF is described in a *Resource Catalog* by specifications, as depicted in Figure 3, where a distributed AMF is implemented as a Kubernetes *StatefulSet* (represented as *SoftwareSpecification*) containing Pods (also represented as *SoftwareSpecification*). An AMF node (represented as *ResourceFunction*) implements interface connection points (represented as *APISpecification*) for internal RPC and external N2 (and normally many others).

Figure 4 depicts the *Function* of an example distributed AMF instance, described in a *Resource Inventory*. The distributed AMF instance (represented as composite *ResourceFunction*) includes a c*onnectionPoint* attribute referencing the N2 interfaces (represented as *API*) while the *connectivity* attribute may contain a topology graph describing the internal nodes (represented as atomic *ResourceFunction*) connected through the RPC interfaces (represented as *API*).

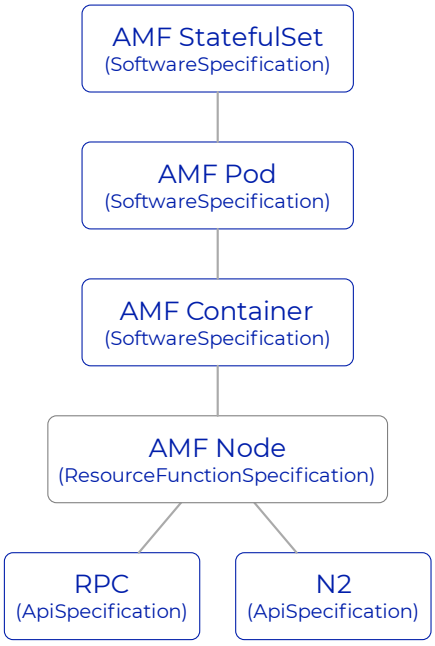


Figure 3: 5GC AMF Specification

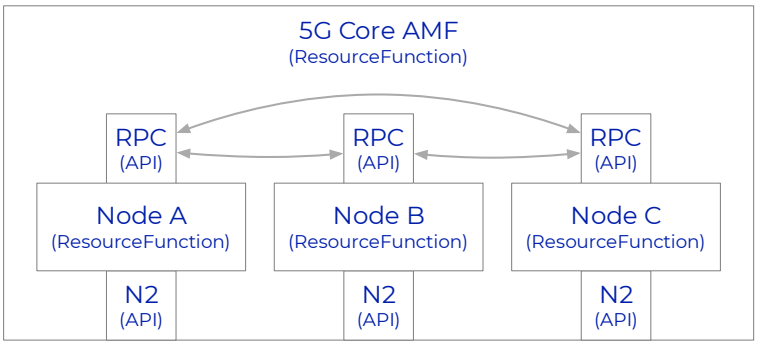


Figure 4: Distributed AMF

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ODA Components

TM Forum's own *Open Digital Architecture (ODA)* is composed of *ODA Components* which are implemented by *Software* and realized as *Processes* making *Functions* available over *APIs*.

In Figure 5 a simplified example of a vendor (Acme) provided *ODA Component* instance is depicted. In this contrived example the instantiated *ODA Component* (represented as *InstalledSoftware*) provides a compound *ResourceFunction* (*Acme Component*) composed of vendor specific, and ODA specified, *ResourceFunctions*.

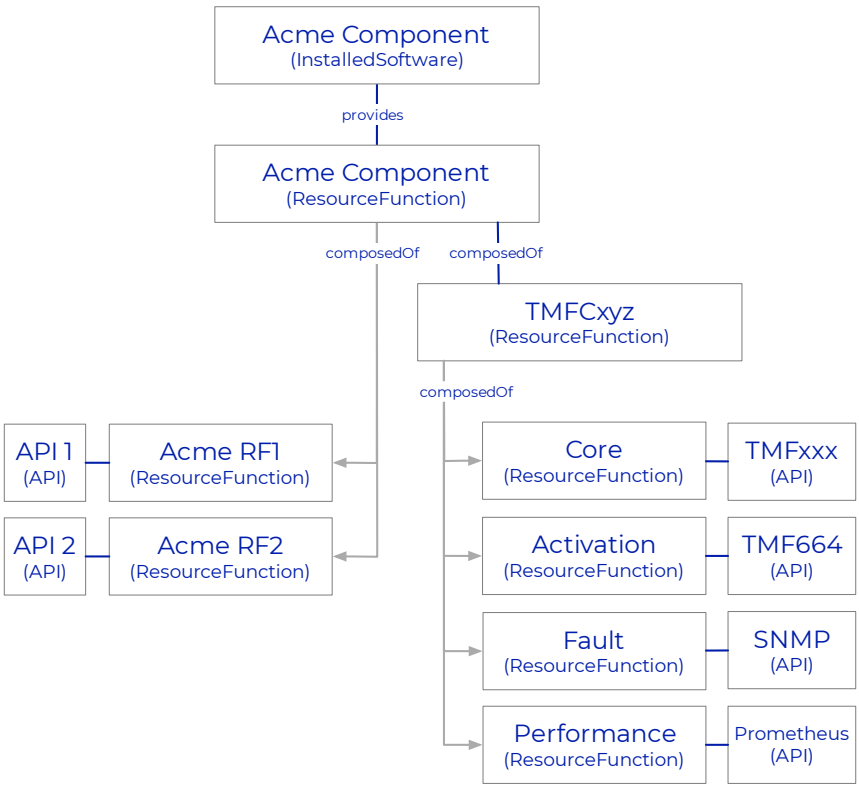


Figure 5: Vendor Provided ODA Component

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance *SoftwareSpecification* and *APISpecification* inheriting properties from the abstract *SoftwareResourceSpecification* entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines v3.0 Part 2 document.

The ***@type*** attribute provides a way to represent the actual class type of an entity. For example, within a list of *ResourceSpecification* instances some may be instances of *SoftwareSpecification*, *ResourceFunctionSpecification*, *APISpecification*, *HostingRequirementSpecification* and *SoftwareSupportPackageSpecification*. The *@type* gives this information. All resources and sub-resources of this API have a *@type* attributes that can be provided when this is useful.

The ***@referredType*** can be used within reference entities (like for instance an *SoftwareSupportPackageRef* object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the *@type*, when used, denotes the class type of the reference itself, such as and not the class type of the referred object. However since reference classes are rarely sub-classed, *@type* is generally not useful in reference objects.

The ***@schemaLocation*** property can be used in resources to allow specifying user-defined properties of an *Entity* or to specify the expected *characteristics* of an entity.

The ***@baseType*** attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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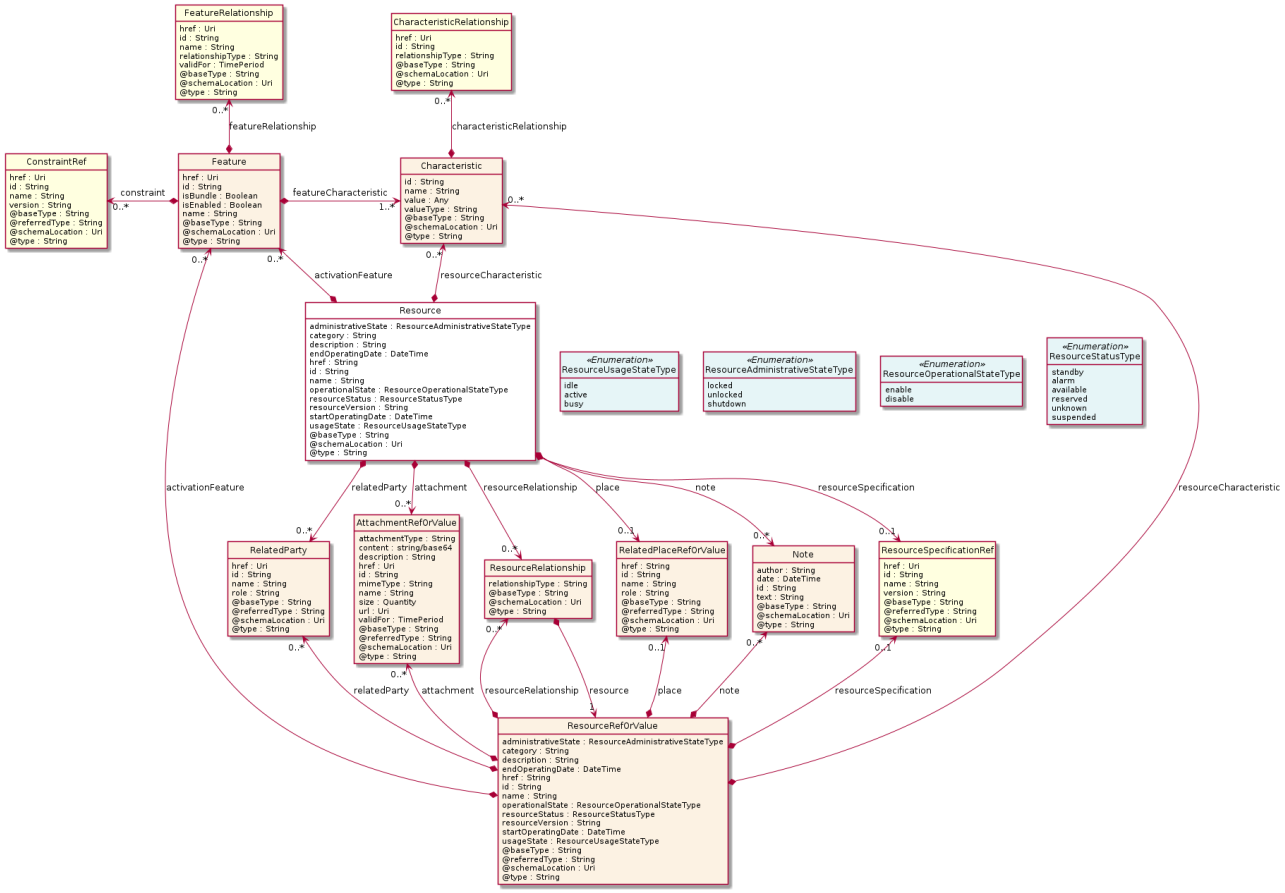
RESOURCE MODEL

Managed Entity and Task Resource Models

Resource resource

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources (e.g. TPE, EQUIPMENT) in the inventory.

Resource model



Field descriptions

Resource fields

href	A string. The URI for the object itself.
id	A string. Identifier of an instance of the resource. Required to be unique within the resource type. Used in URIs as the identifier for specific instances of a type.
@baseType	A string. When sub-classing, this defines the super-class.

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@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
activationFeature	A list of features (Feature [*]). Configuration features.
administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.
endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.
href	A string. The URI for the object itself.
id	A string. Identifier of an instance of the resource. Required to be unique within the resource type. Used in URIs as the identifier for specific instances of a type.
name	A string. A string used to give a name to the resource.
note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource

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	needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.

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@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

Feature sub-resource

Configuration feature.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureCharacteristic	A list of characteristics (Characteristic [1..*]). This is a list of Characteristics for a particular feature.
featureRelationship	A list of feature relationships (FeatureRelationship [*]). Configuration feature.
isBundle	A boolean. True if this is a feature group. Default is false.
isEnabled	A boolean. True if this feature is enabled. Default is true.
name	A string. This is the name for the feature.

FeatureRelationship sub-resource

Configuration feature.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the target feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
name	A string. This is the name of the target feature.
relationshipType	A string. This is the type of the feature relationship.
validFor	A time period. The period for which this feature relationship is valid.

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Note sub-resource

Extra information about a given entity.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.
text	A string. Text of the note.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

RelatedPlaceRefOrValue sub-resource

Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.

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@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

ResourceRefOrValue sub-resource

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the related ResourceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A string used to give a name to the resource.
href	A string. The URI for the object itself.
id	A string. Identifier of an instance of the resource. Required to be unique within the resource type. Used in URIs as the identifier for specific instances of a type.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.
endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.

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startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.
activationFeature	A list of features (Feature [*]). Configuration features.
administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.
note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set

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	of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

ResourceRelationship sub-resource

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g.: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A resource ref or value (ResourceRefOrValue). Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the related ResourceRefOrValue class itself.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.

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@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

ResourceSpecificationRef relationship

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components.
A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Resource Specification version.

Json representation sample

Provided below is the json representation of an example of a 'Resource' resource object


```
{
  "id": "6381aee0-6e5a-436f-8e20-b231d3efd712",
  "href": "/resourceInventory/v4/resource/6381aee0-6e5a-436f-8e20-b231d3efd712",
  "name": "Encabulator Software",
  "description": "This software realizes an instance of Encabulator",
  "@type": "InstalledSoftware",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "lastUpdate": "2022-02-04T15:10:05Z",
  "category": "Software",
```


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```
  "resourceSpecification": {
    "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "name": "Encabulator Software Specification",
    "@referredType": "SoftwareSpecification"
  },
  "isUTCTime": true,
  "lastStartTime": "2022-02-04T15:10:01Z",
  "targetPlatform": "server",
  "numProcessesActiveCurrent": 1,
  "numUsersCurrent": 0,
  "pagingFileSizeCurrent": {
    "amount": 49152,
    "units": "4 KiB"
  },
  "processMemorySizeCurrent": {
    "amount": 1399714284,
    "units": "bytes"
  },
  "swapSpaceUsedCurrent": {
    "amount": 0,
    "units": "bytes"
  },
  "serialNumber": "831038429801",
  "resourceRelationship": [
    {
      "relationshipType": "uses",
      "resource": {
        "id": "a1976344-3933-4942-874a-ded80c1e5d83",
        "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
        "name": "Encabulator Platform Requirement",
        "@referredType": "HostingPlatformRequirement"
      }
    },
    {
      "relationshipType": "provides",
      "resource": {
        "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "name": "Encabulator Function",
        "@referredType": "ResourceFunction"
      }
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "OperatingSystem",
      "value": "Debian 11 (bullseye)"
    }
  ]
}
```

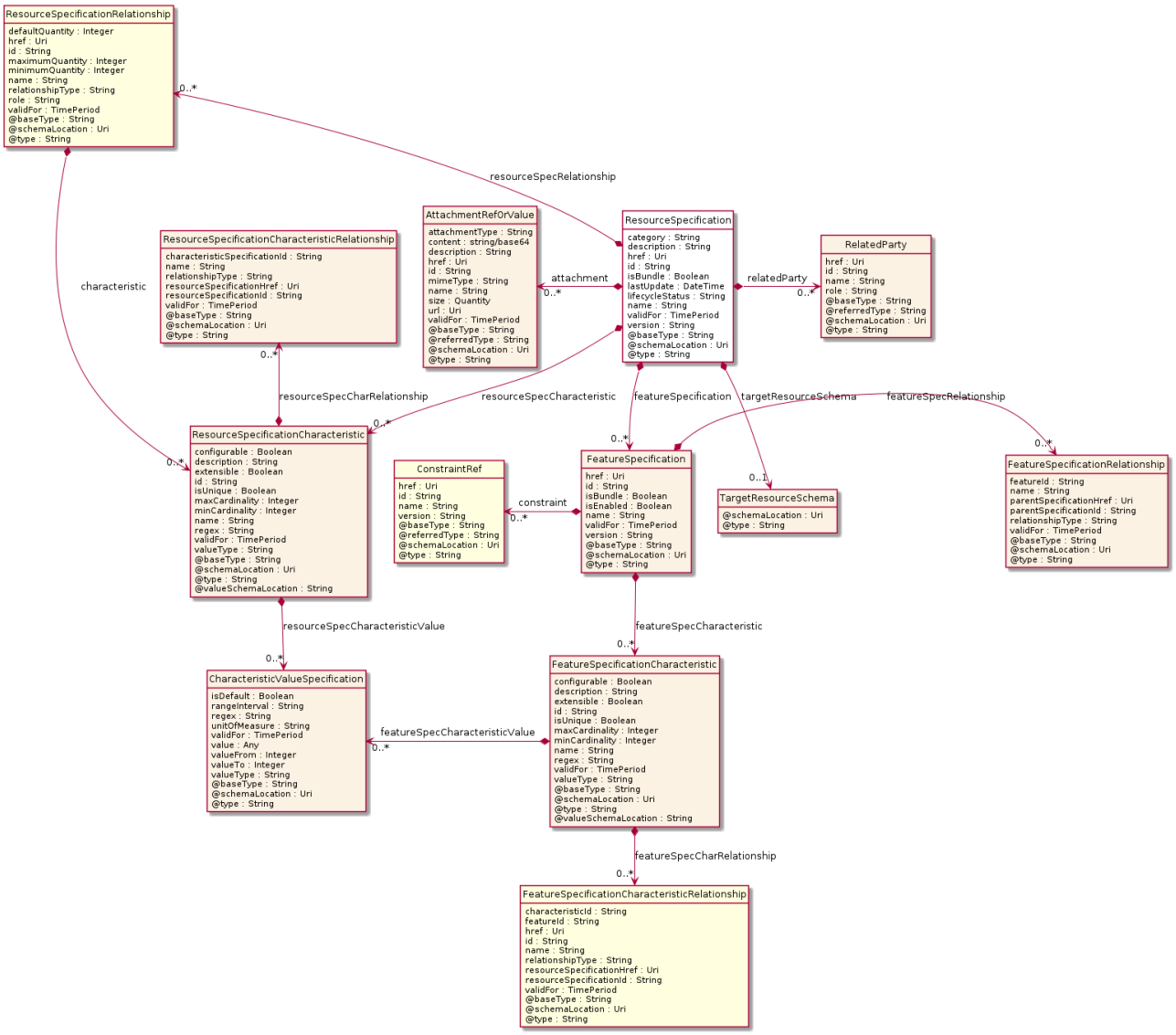

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Resource Specification resource

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components.
A ResourceSpecification is a base class that represents a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

Resource model



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Field descriptions

ResourceSpecification fields

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
href	An uri (Uri). Hyperlink reference to this REST resource.
id	A string. Unique identifier of this REST resource.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). Complements the description of an element (for instance a resource) through video, pictures ...
category	A string. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A string. Description of this REST resource.
featureSpecification	A list of feature specifications (FeatureSpecification [*]). A list of Features for this specification.
href	An uri (Uri). Hyperlink reference to this REST resource.
id	A string. Unique identifier of this REST resource.
isBundle	A boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).
lastUpdate	A date time (DateTime). Date and time of the last update of this REST resource.
lifecycleStatus	A string. Used to indicate the current lifecycle status of the resource specification.
name	A string. Name given to this REST resource.
relatedParty	A list of related parties (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
resourceSpecCharacteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic quality or distinctive feature of a ResourceSpecification. The characteristic can be taken on a discrete value, such as color, can take on a range of values, (for example, sensitivity of 100-240 mV), or can be derived from a formula (for example, usage time (hrs) = 30 - talk time *3). Certain characteristics,

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	such as color, may be configured during the ordering or some other process.
resourceSpecRelationship	A list of resource specification relationships (ResourceSpecificationRelationship [*]). A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
targetResourceSchema	A target resource schema (TargetResourceSchema). A target resource schema reference. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A time period. The period for which this REST resource is valid.
version	A string. Resource Specification version.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

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CharacteristicValueSpecification sub-resource

specification of a value (number or text or an object) that can be assigned to a Characteristic.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
isDefault	A boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A string. A regular JsonLdExpression constraint for given value.
unitOfMeasure	A string. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A time period. The period of time for which a value is applicable.
value	And any (Any). A discrete value that the characteristic can take on, or the actual value of the characteristic.
valueFrom	An integer. The low range value that a characteristic can take on.
valueTo	An integer. The upper range value that a characteristic can take on.
valueType	A string. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

FeatureSpecification sub-resource

Specification for resource, service or product features.

href	An uri (Uri). Hyperlink reference.
id	A string. Identifier of the feature specification. Must be locally unique within the containing specification, thus allowing direct access to the feature spec.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureSpecCharacteristic	A list of feature specification characteristics (FeatureSpecificationCharacteristic [*]). This is a list of characteristics for a particular feature.
featureSpecRelationship	A list of feature specification relationships (FeatureSpecificationRelationship [*]). A dependency, exclusivity or aggregation relationship between/among feature specifications.
isBundle	A boolean. A flag indicating if this is a feature group (true) or not (false).
isEnabled	A boolean. A flag indicating if the feature is enabled (true) or not (false).
name	A string. Unique name given to the feature specification.
validFor	A time period. The period for which this feature specification is valid.
version	A string. Version of the feature specification.

FeatureSpecificationCharacteristic sub-resource

Configuration feature characteristic specification.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.

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minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureSpecCharRelationship	A list of feature specification characteristic relationships (FeatureSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among feature characteristics.
featureSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). Used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a FeatureCharacteristic object.
validFor	A time period. The period of time for which a characteristic is applicable.

FeatureSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among FeatureSpecificationCharacteristics.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

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characteristicId	A string. Unique identifier of the characteristic within the target feature specification.
featureId	A string. Unique identifier of the target feature specification within the resource specification.
name	A string. Name of the target characteristic.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target feature and feature characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target feature and feature characteristic.
validFor	A time period. The period for which the object is valid.

FeatureSpecificationRelationship sub-resource

Relationship between feature specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureId	A string. Unique identifier of the target feature specification.
name	A string. This is the name of the target feature specification.
parentSpecificationHref	An uri (Uri). Hyperlink reference to the parent specification containing the target feature.
parentSpecificationId	A string. Unique identifier of the parent specification containing the target feature.
relationshipType	A string. This is the type of the feature specification relationship.
validFor	A time period. The period for which this feature specification relationship is valid.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

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RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

ResourceSpecificationCharacteristic sub-resource

This class defines the characteristic features of a resource specification. Every ResourceSpecification has a variety of important attributes, methods, constraints, and
relationships, which distinguish a resource specification from other resource specifications.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone

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	numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
resourceSpecCharRelationship	A list of resource specification characteristic relationships (ResourceSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
resourceSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a ResourceSpecificationCharacteristic object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding ResourceSpecificationCharacteristic object can take on.
validFor	A time period. The period of time for which a characteristic is applicable.

ResourceSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among ResourceSpecificationCharacteristics. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.

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@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicSpecificationId	A string. Unique identifier of the characteristic within the specification.
name	A string. Name of the target characteristic within the specification.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target characteristic.
validFor	A time period. The period for which the object is valid.

ResourceSpecificationRelationship sub-resource

A migration, substitution, dependency or exclusivity relationship between/among resource specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic that refines the relationship. For example, consider the relationship between a slot and a card. For a half-height card it is important to know the position at which the card is inserted, so a characteristic Position might be defined on the relationship to allow capturing of this in the inventory.
defaultQuantity	An integer. The default number of the related resource that should be instantiated, for example a rack would typically have 4 cards, although it could support more.
href	An uri (Uri). Reference of the target ResourceSpecification.
id	A string. Unique identifier of target ResourceSpecification.
maximumQuantity	An integer. The maximum number of the related resource that should be instantiated, for example a rack supports a maximum of 16 cards.

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minimumQuantity	An integer. The minimum number of the related resource that should be instantiated, for example a rack must have at least 1 card.
name	A string. The name given to the target resource specification instance.
relationshipType	A string. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A string. The association role for this resource specification.
validFor	A time period. The period for which the ResourceSpecRelationship is valid.

TargetResourceSchema sub-resource

The reference object to the schema and type of target resource which is described by resource specification.

@schemaLocation	An uri (Uri). This field provides a link to the schema describing the target resource.
@type	A string. Class type of the target resource.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

Json representation sample

Provided below is the json representation of an example of a 'ResourceSpecification' resource object


```
{
  "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
  "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
  "name": "Encabulator Software Specification",
  "description": "This specification describes the Encabulator software",
```


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```
"@type": "SoftwareSpecification",
"validFor": {
   "startDateTime": "2022-11-28T08:00:00Z",
   "endDateTime": "2023-11-29T00:00:00Z"
},
"lastUpdate": "2022-02-03T07:59:59Z",
"lifecycleStatus": "Active",
"isBundle": false,
"category": "Software",
"targetResourceSchema": {
   "@type": "InstalledSoftware"
},
"relatedParty": [
   {
      "id": "ab5660aa-064d-440c-b7d1-b943e42b904e",
      "href": "/partyManagement/v4/organization/ab5660aa-064d-440c-b7d1-b943e42b904e",
      "role": "supplier",
      "name": "General Electric"
   }
],
"releaseStatus": "firstCustomerShipment",
"buildNumber": "encabulator-1.0.0",
"majorVersion": "1",
"minorVersion": "0",
"maintenanceVersion": "0",
"otherDesignator": "alpha",
"isDistributable": true,
"installSize": {
   "amount": 9143256,
   "units": "bytes"
},
"isExperimental": false,
"numUsersMax": 8,
"numberProcessActiveTotal": 0,
"resourceSpecRelationship": [
   {
      "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "name": "Encabulator Platform Requirement Specification",
      "relationshipType": "requires"
   },
   {
      "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "name": "Encabulator Function Specification",
      "relationshipType": "provides"
   }
],
"softwareSupportPackage": {
   "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
   "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
   "name": "Encabulator Software Package",
   "@referredType": "SoftwareSupportPackage"
},
"resourceSpecCharacteristic": [
```


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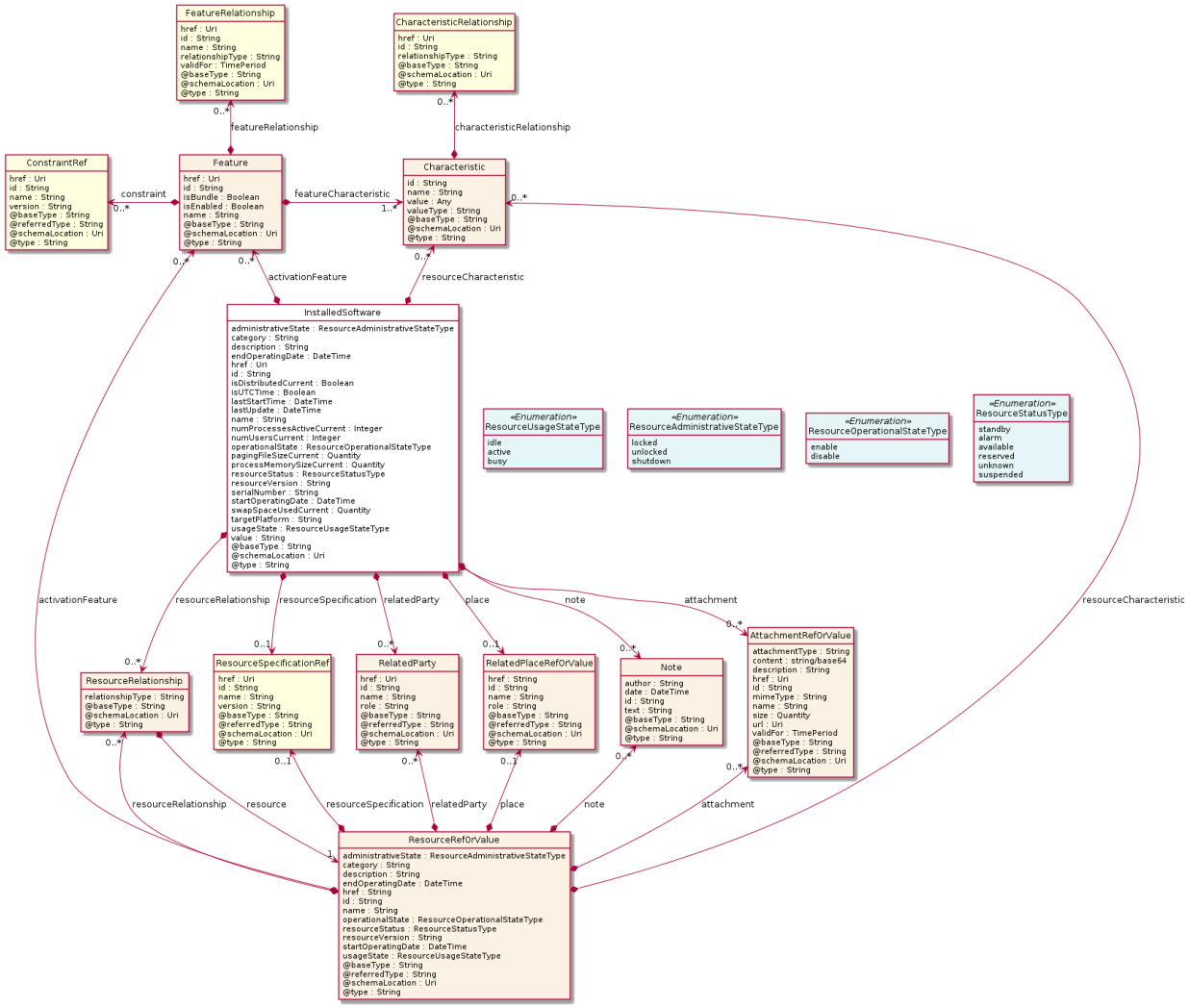
```
    {
      "name": "OperatingSystem",
      "description": "Operating System",
      "valueType": "String",
      "configurable": true,
      "@type": "ResourceSpecificationCharacteristic",
      "minCardinality": 1,
      "maxCardinality": 3,
      "isUnique": true,
      "extensible": false,
      "resourceSpecCharacteristicValue": [
        {
          "isDefault": true,
          "value": "Debian 11 (bullseye)"
        },
        {
          "isDefault": false,
          "value": "Ubuntu 20.04 LTS (focal)"
        },
        {
          "isDefault": false,
          "value": "RHEL 9"
        }
      ]
    }
  ]
}
```


Installed Software resource

A SoftwareSpecification deployed using the SoftwareSupportPackage on platform which meets the HostingPlatformRequirements.

Resource model

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Field descriptions

InstalledSoftware fields

isDistributedCurrent	A boolean. True defines this SoftwareResource as being currently distributed across several Resource nodes.
lastUpdate	A date time (DateTime). The time of the last update of this resource.
targetPlatform	A string. Defines the target platform that the SoftwareResource will be deployed on.
value	A string. the value of the logical resource. E.g. '0746712345' for MSISDN's.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.

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endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
name	A string. A string used to give a name to the resource.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
isUTCTime	A boolean. Indicates whether this software uses absolute (UTC) time.
lastStartTime	A date time (DateTime). The time of the last start of this software.
numProcessesActiveCurrent	An integer. The current number of total processes that the Software can support at any given time.
numUsersCurrent	An integer. The current number of users of the Software.
pagingFileSizeCurrent	A quantity (Quantity). Defines the current number, and size of, stored paging files for the software. If this instance doesn't support paging files, then the value of this attribute should be 0.
processMemorySizeCurrent	A quantity (Quantity). The amount of memory currently allocated to a process.
serialNumber	A string. A set of identifying characters and/or numbers assigned to, and usually marked on, each of a series of identical Software instances.
swapSpaceUsedCurrent	A quantity (Quantity). The amount of swap space currently in use. If the Software doesn't distinguish swap space from space used for paging files, then the value of this attribute should be 0.
activationFeature	A list of features (Feature [*]). Configuration features.

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administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.
note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and

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	relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.

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@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

Feature sub-resource

Configuration feature.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.

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featureCharacteristic	A list of characteristics (Characteristic [1..*]). This is a list of Characteristics for a particular feature.
featureRelationship	A list of feature relationships (FeatureRelationship [*]). Configuration feature.
isBundle	A boolean. True if this is a feature group. Default is false.
isEnabled	A boolean. True if this feature is enabled. Default is true.
name	A string. This is the name for the feature.

FeatureRelationship sub-resource

Configuration feature.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the target feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
name	A string. This is the name of the target feature.
relationshipType	A string. This is the type of the feature relationship.
validFor	A time period. The period for which this feature relationship is valid.

Note sub-resource

Extra information about a given entity.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.
text	A string. Text of the note.

Quantity sub-resource

An amount in a given unit.

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amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

RelatedPlaceRefOrValue sub-resource

Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

ResourceRefOrValue sub-resource

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Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the related ResourceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A string used to give a name to the resource.
href	A string. The URI for the object itself.
id	A string. Identifier of an instance of the resource. Required to be unique within the resource type. Used in URIs as the identifier for specific instances of a type.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.
endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.
activationFeature	A list of features (Feature [*]). Configuration features.
administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.
note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value

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	linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

ResourceRelationship sub-resource

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

@baseType	A string. When sub-classing, this defines the super-class.

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@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g.: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A resource ref or value (ResourceRefOrValue). Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the related ResourceRefOrValue class itself.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

ResourceSpecificationRef relationship

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components.
A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Resource Specification version.

Json representation sample

Provided below is the json representation of an example of a 'InstalledSoftware' resource object


```
{
  "id": "6381aee0-6e5a-436f-8e20-b231d3efd712",
  "href": "/resourceInventory/v4/resource/6381aee0-6e5a-436f-8e20-b231d3efd712",
  "name": "Encabulator Software",
  "description": "This software realizes an instance of Encabulator",
  "@type": "InstalledSoftware",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "lastUpdate": "2022-02-04T15:10:05Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "name": "Encabulator Software Specification",
    "@referredType": "SoftwareSpecification"
  },
  "isUTCTime": true,
  "lastStartTime": "2022-02-04T15:10:01Z",
  "targetPlatform": "server",
  "numProcessesActiveCurrent": 1,
  "numUsersCurrent": 0,
  "pagingFileSizeCurrent": {
    "amount": 49152,
    "units": "4 KiB"
  },
  "processMemorySizeCurrent": {
    "amount": 1399714284,
    "units": "bytes"
  },
  "swapSpaceUsedCurrent": {
    "amount": 0,
    "units": "bytes"
```


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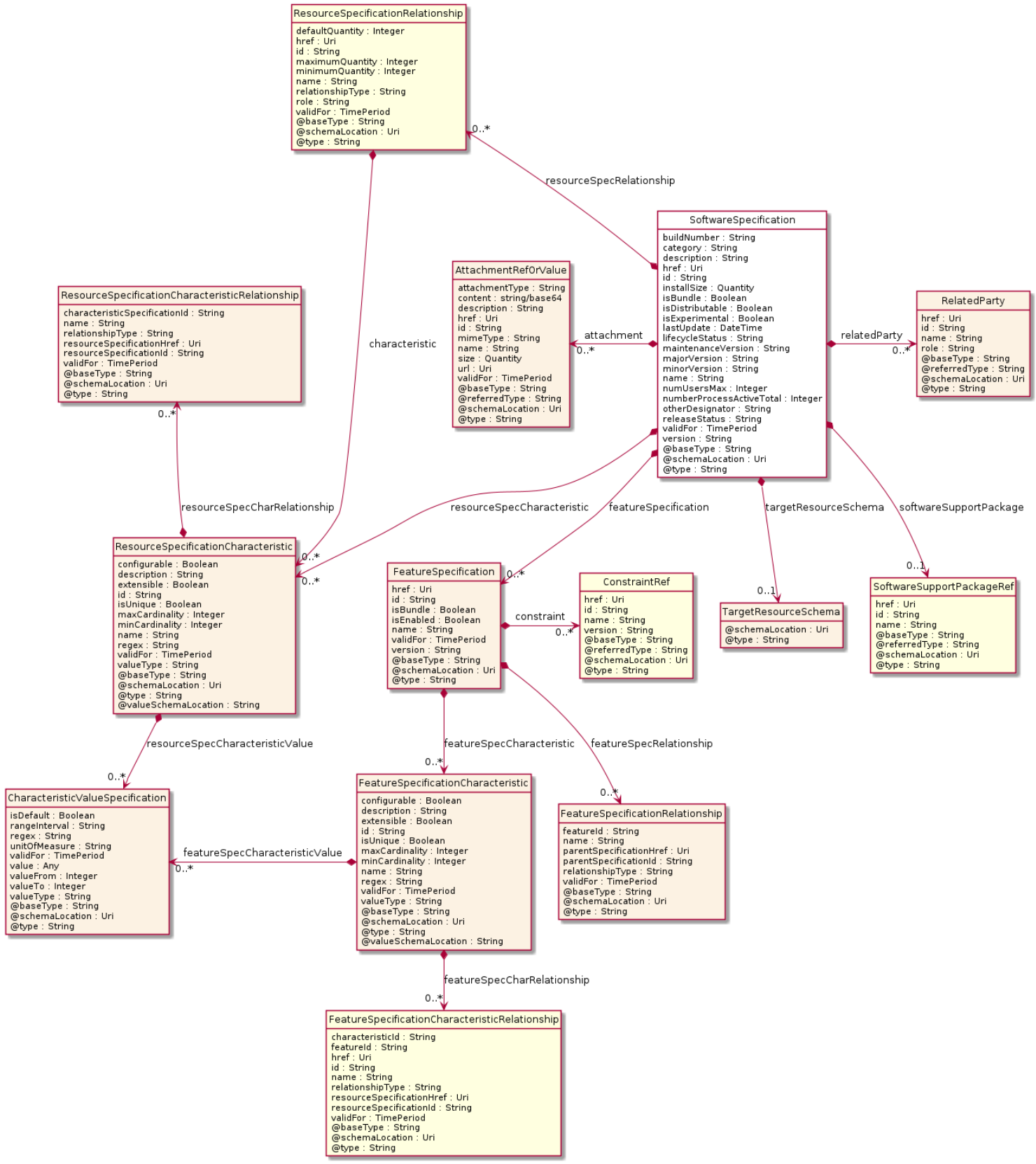
```
  },
  "serialNumber": "831038429801",
  "resourceRelationship": [
    {
      "relationshipType": "uses",
      "resource": {
        "id": "a1976344-3933-4942-874a-ded80c1e5d83",
        "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
        "name": "Encabulator Platform Requirement",
        "@referredType": "HostingPlatformRequirement"
      }
    },
    {
      "relationshipType": "provides",
      "resource": {
        "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "name": "Encabulator Function",
        "@referredType": "ResourceFunction"
      }
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "OperatingSystem",
      "value": "Debian 11 (bullseye)"
    }
  ]
}
```


Software Specification resource

A base class is used to define the invariant characteristics and behavior (attributes, constraints, and relationships) of an InstalledSoftware.

Resource model

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Field descriptions

SoftwareSpecification fields

buildNumber	A string. Defines the internal identifier for the compilation of this unit of a SoftwareResourceSpecification.
isDistributable	A boolean. Designates the SoftwareResource as being able to be distributed across multiple physically distinct Resources.
isExperimental	A boolean. True represents a special build (perhaps as a trial to suit specific customer needs).
maintenanceVersion	A string. Defines the maintenance number component of the version of SoftwareResourceSpecification.
majorVersion	A string. Defines the major number component of the version of SoftwareResourceSpecification.
minorVersion	A string. Defines the minor number component of the version of SoftwareResourceSpecification.
otherDesignator	A string. Enables the manufacturer to label this software as having some set of special restrictions.
releaseStatus	A string. Defines the particular release status of this SoftwareResourceSpecification.
category	A string. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A string. Description of this REST resource.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
isBundle	A boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).
lastUpdate	A date time (DateTime). Date and time of the last update of this REST resource.
lifecycleStatus	A string. Used to indicate the current lifecycle status of the resource specification.
name	A string. Name given to this REST resource.
version	A string. Resource Specification version.
@baseType	A string. When sub-classing, this defines the super-class.

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@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
numUsersMax	An integer. Defines the maximum number of users for a SoftwareSpecifciation.
numberProcessActiveTotal	An integer. Defines the maximum number of total processes that a SoftwareSpecifciation can support at any given time.
softwareSupportPackage	A software support package reference (SoftwareSupportPackageRef). Reference to a SoftwareSupportPackage.
installSize	A quantity (Quantity). The amount and units (e.g. 800MB) required to install the SoftwareResource.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). Complements the description of an element (for instance a resource) through video, pictures ...
featureSpecification	A list of feature specifications (FeatureSpecification [*]). A list of Features for this specification.
relatedParty	A list of related parties (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
resourceSpecCharacteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic quality or distinctive feature of a ResourceSpecification. The characteristic can be taken on a discrete value, such as color, can take on a range of values, (for example, sensitivity of 100-240 mV), or can be derived from a formula (for example, usage time (hrs) = 30 - talk time *3). Certain characteristics, such as color, may be configured during the ordering or some other process.
resourceSpecRelationship	A list of resource specification relationships (ResourceSpecificationRelationship [*]). A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
targetResourceSchema	A target resource schema (TargetResourceSchema). A target resource schema reference. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A time period. The period for which this REST resource is valid.

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AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

CharacteristicValueSpecification sub-resource

specification of a value (number or text or an object) that can be assigned to a Characteristic.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
isDefault	A boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.

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rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A string. A regular JsonLdExpression constraint for given value.
unitOfMeasure	A string. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A time period. The period of time for which a value is applicable.
value	An any (Any). A discrete value that the characteristic can take on, or the actual value of the characteristic.
valueFrom	An integer. The low range value that a characteristic can take on.
valueTo	An integer. The upper range value that a characteristic can take on.
valueType	A string. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

FeatureSpecification sub-resource

Specification for resource, service or product features.

href	An uri (Uri). Hyperlink reference.
id	A string. Identifier of the feature specification. Must be locally unique within the containing specification, thus allowing direct access to the feature spec.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureSpecCharacteristic	A list of feature specification characteristics (FeatureSpecificationCharacteristic [*]). This is a list of characteristics for a particular feature.
featureSpecRelationship	A list of feature specification relationships (FeatureSpecificationRelationship [*]). A dependency, exclusivity or aggregation relationship between/among feature specifications.

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isBundle	A boolean. A flag indicating if this is a feature group (true) or not (false).
isEnabled	A boolean. A flag indicating if the feature is enabled (true) or not (false).
name	A string. Unique name given to the feature specification.
validFor	A time period. The period for which this feature specification is valid.
version	A string. Version of the feature specification.

FeatureSpecificationCharacteristic sub-resource

Configuration feature characteristic specification.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.

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valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureSpecCharRelationship	A list of feature specification characteristic relationships (FeatureSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among feature characteristics.
featureSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). Used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a FeatureCharacteristic object.
validFor	A time period. The period of time for which a characteristic is applicable.

FeatureSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among FeatureSpecificationCharacteristics.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicId	A string. Unique identifier of the characteristic within the target feature specification.
featureId	A string. Unique identifier of the target feature specification within the resource specification.
name	A string. Name of the target characteristic.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.

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resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target feature and feature characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target feature and feature characteristic.
validFor	A time period. The period for which the object is valid.

FeatureSpecificationRelationship sub-resource

Relationship between feature specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureId	A string. Unique identifier of the target feature specification.
name	A string. This is the name of the target feature specification.
parentSpecificationHref	An uri (Uri). Hyperlink reference to the parent specification containing the target feature.
parentSpecificationId	A string. Unique identifier of the parent specification containing the target feature.
relationshipType	A string. This is the type of the feature specification relationship.
validFor	A time period. The period for which this feature specification relationship is valid.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.

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id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

ResourceSpecificationCharacteristic sub-resource

This class defines the characteristic features of a resource specification. Every ResourceSpecification has a variety of important attributes, methods, constraints, and
relationships, which distinguish a resource specification from other resource specifications.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.

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name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
resourceSpecCharRelationship	A list of resource specification characteristic relationships (ResourceSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
resourceSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a ResourceSpecificationCharacteristic object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding ResourceSpecificationCharacteristic object can take on.
validFor	A time period. The period of time for which a characteristic is applicable.

ResourceSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among ResourceSpecificationCharacteristics. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicSpecificationId	A string. Unique identifier of the characteristic within the specification.

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name	A string. Name of the target characteristic within the specification.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target characteristic.
validFor	A time period. The period for which the object is valid.

ResourceSpecificationRelationship sub-resource

A migration, substitution, dependency or exclusivity relationship between/among resource specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic that refines the relationship. For example, consider the relationship between a slot and a card. For a half-height card it is important to know the position at which the card is inserted, so a characteristic Position might be defined on the relationship to allow capturing of this in the inventory.
defaultQuantity	An integer. The default number of the related resource that should be instantiated, for example a rack would typically have 4 cards, although it could support more.
href	An uri (Uri). Reference of the target ResourceSpecification.
id	A string. Unique identifier of target ResourceSpecification.
maximumQuantity	An integer. The maximum number of the related resource that should be instantiated, for example a rack supports a maximum of 16 cards.
minimumQuantity	An integer. The minimum number of the related resource that should be instantiated, for example a rack must have at least 1 card.
name	A string. The name given to the target resource specification instance.
relationshipType	A string. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A string. The association role for this resource specification.

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validFor	A time period. The period for which the ResourceSpecRelationship is valid.

TargetResourceSchema sub-resource

The reference object to the schema and type of target resource which is described by resource specification.

@schemaLocation	An uri (Uri). This field provides a link to the schema describing the target resource.
@type	A string. Class type of the target resource.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

SoftwareSupportPackageRef relationship

Reference to a SoftwareSupportPackage.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

Provided below is the json representation of an example of a 'SoftwareSpecification' resource object


```
{
  "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
  "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
  "name": "Encabulator Software Specification",
  "description": "This specification describes the Encabulator software",
  "@type": "SoftwareSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lastUpdate": "2022-02-03T07:59:59Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "InstalledSoftware"
  },
  "relatedParty": [
    {
      "id": "ab5660aa-064d-440c-b7d1-b943e42b904e",
      "href": "/partyManagement/v4/organization/ab5660aa-064d-440c-b7d1-b943e42b904e",
      "role": "supplier",
      "name": "General Electric"
    }
  ],
  "releaseStatus": "firstCustomerShipment",
  "buildNumber": "encabulator-1.0.0",
  "majorVersion": "1",
  "minorVersion": "0",
  "maintenanceVersion": "0",
  "otherDesignator": "alpha",
  "isDistributable": true,
  "installSize": {
    "amount": 9143256,
    "units": "bytes"
  },
  "isExperimental": false,
  "numUsersMax": 8,
  "numberProcessActiveTotal": 0,
  "resourceSpecRelationship": [
    {
      "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "name": "Encabulator Platform Requirement Specification",
      "relationshipType": "requires"
    },
    {
      "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
```


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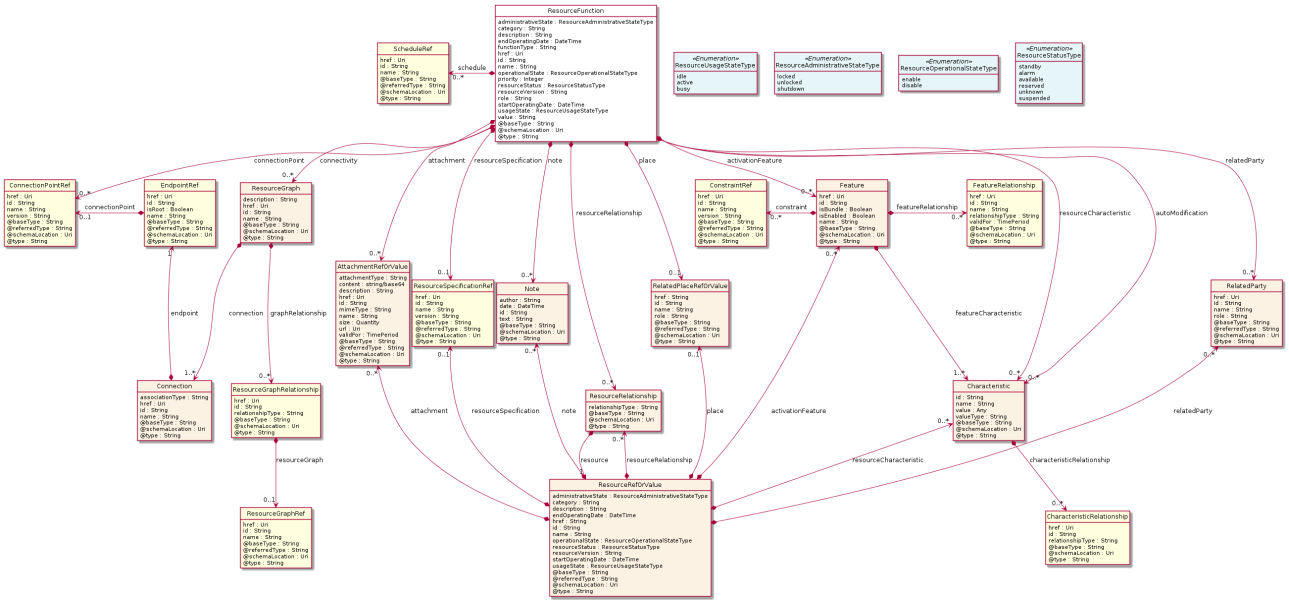
```
        "name": "Encabulator Function Specification",
        "relationshipType": "provides"
      }
    ],
    "softwareSupportPackage": {
      "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
      "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
      "name": "Encabulator Software Package",
      "@referredType": "SoftwareSupportPackage"
    },
    "resourceSpecCharacteristic": [
      {
        "name": "OperatingSystem",
        "description": "Operating System",
        "valueType": "String",
        "configurable": true,
        "@type": "ResourceSpecificationCharacteristic",
        "minCardinality": 1,
        "maxCardinality": 3,
        "isUnique": true,
        "extensible": false,
        "resourceSpecCharacteristicValue": [
          {
            "isDefault": true,
            "value": "Debian 11 (bullseye)"
          },
          {
            "isDefault": false,
            "value": "Ubuntu 20.04 LTS (focal)"
          },
          {
            "isDefault": false,
            "value": "RHEL 9"
          }
        ]
      }
    ]
}
```


Resource Function resource

A ResourceFunction is a behavior to transform inputs of any nature into outputs of any nature independently from the way it is provided.

Resource model

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Field descriptions

ResourceFunction fields

value	A string. the value of the logical resource. E.g. '0746712345' for MSISDN's.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.
endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
name	A string. A string used to give a name to the resource.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A string. When sub-classing, this defines the sub-class Extensible name.
autoModification	A list of characteristics (Characteristic [*]). List of the kinds of auto-modifications that are applied to a given network service e.g. what can be scaled.
connectionPoint	A list of connection point references (ConnectionPointRef [*]). External connection points of the resource function. These are the service access points (SAP) where inputs and outputs of the function are available.
connectivity	A list of resource graphs (ResourceGraph [*]). Internal connectivity of contained resource functions.
functionType	A string. A type of the Resource Function as specified by the provider of the API.
priority	An integer. Priority of the Resource Function. Decides what happens in a contention scenario.
role	A string. Role of the Resource Function. Used when Resource Function is a component of a composite Resource Function and the exact role of the service within the composite is not clear from descriptor/location.
schedule	A list of schedule references (ScheduleRef [*]). This is a reference to a schedule. Allows consumers to schedule modifications to the service at certain times.
activationFeature	A list of features (Feature [*]). Configuration features.
administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.
note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.

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resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.

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@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.

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@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

Connection sub-resource

An edge in a resource graph.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier for graph edge.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
associationType	A string. Association type.
endpoint	An endpoint reference (EndpointRef). Resource graph vertices connected by this edge.
name	A string. Descriptive name for graph edge.

Feature sub-resource

Configuration feature.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureCharacteristic	A list of characteristics (Characteristic [1..*]). This is a list of Characteristics for a particular feature.
featureRelationship	A list of feature relationships (FeatureRelationship [*]). Configuration feature.

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isBundle	A boolean. True if this is a feature group. Default is false.
isEnabled	A boolean. True if this feature is enabled. Default is true.
name	A string. This is the name for the feature.

FeatureRelationship sub-resource

Configuration feature.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the target feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
name	A string. This is the name of the target feature.
relationshipType	A string. This is the type of the feature relationship.
validFor	A time period. The period for which this feature relationship is valid.

Note sub-resource

Extra information about a given entity.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.
text	A string. Text of the note.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

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RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

RelatedPlaceRefOrValue sub-resource

Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

ResourceGraph sub-resource

Resource graph.

href	An uri (Uri). Hyperlink reference.

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id	A string. Unique identifier of the resource graph.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
connection	A list of connections (Connection [1..*]). Resource graph edges.
description	A string. Description of the resource graph.
graphRelationship	A list of resource graph relationships (ResourceGraphRelationship [*]). Relationships to other resource graphs.
name	A string. Descriptive name for the resource graph.

ResourceGraphRelationship sub-resource

Describes links between resource graphs.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. Semantic of the relationship.
resourceGraph	A resource graph reference (ResourceGraphRef). A reference to the related resource graph.

ResourceRefOrValue sub-resource

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the related ResourceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A string used to give a name to the resource.
href	A string. The URI for the object itself.

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id	A string. Identifier of an instance of the resource. Required to be unique within the resource type. Used in URIs as the identifier for specific instances of a type.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.
endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.
activationFeature	A list of features (Feature [*]). Configuration features.
administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.
note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.

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resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

ResourceRelationship sub-resource

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g.: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

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resource	A resource ref or value (ResourceRefOrValue). Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the related ResourceRefOrValue class itself.

ConnectionPointRef relationship

Connection point reference. A connection point provides a service access point (SAP) for input and/or output of the resource function.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Connection point version.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

EndpointRef relationship

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A vertex in a resource graph.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
connectionPoint	A connection point reference (ConnectionPointRef). The connection point (SAP) of the associated resource function (optional).
isRoot	A boolean. Directionality: true when endpoint is a source, false when a sink. If true for all endpoints connectivity is bidirectional. Default is true.

ResourceGraphRef relationship

Resource graph reference.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRef relationship

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components.
A ResourceSpecification is an abstract base class for representing a generic means for implementing a

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particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Resource Specification version.

ScheduleRef relationship

Schedule reference.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

Provided below is the json representation of an example of a 'ResourceFunction' resource object


```
{
  "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
  "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
  "name": "Encabulator Function",
  "description": "The Encabulator function.",
  "@type": "ResourceFunction",
  "startOperatingDate": "2022-11-28T08:00:00Z",
```

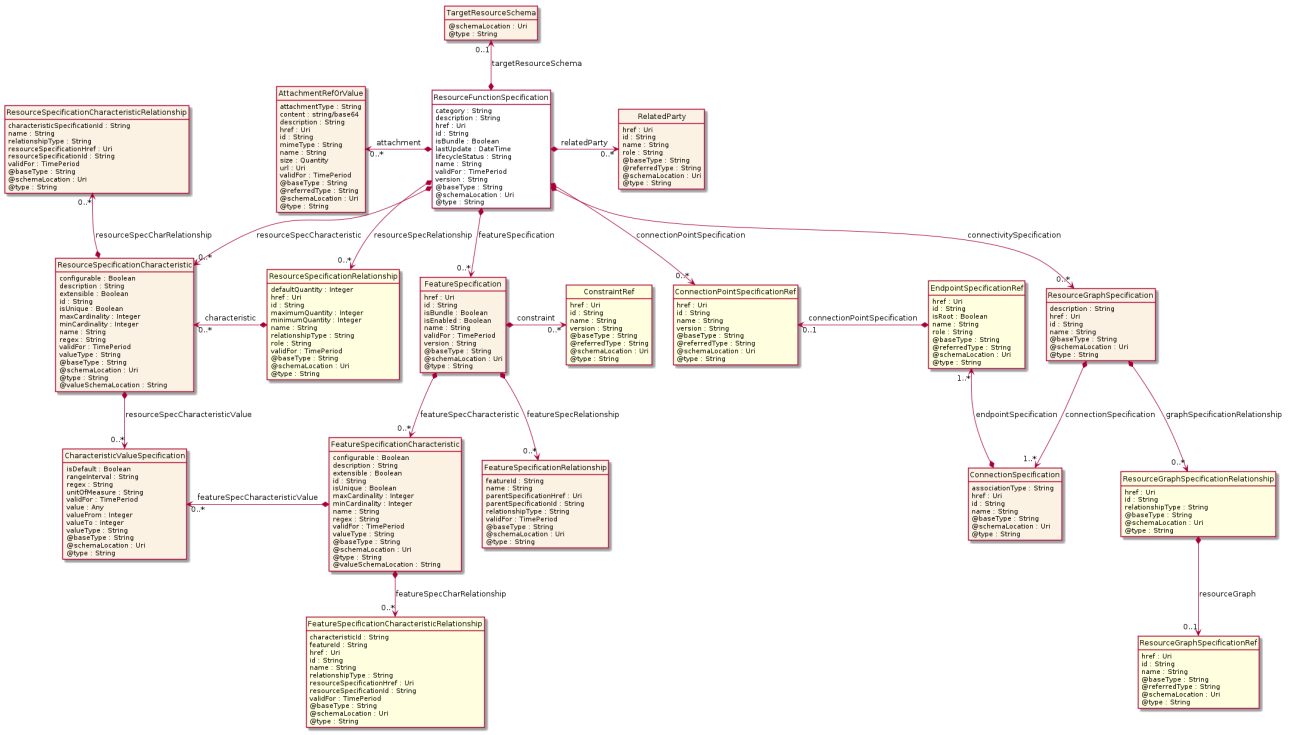

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```
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "name": "Encabulator Function Specification",
    "@referredType": "ResourceFunctionSpecification"
  },
  "resourceRelationship": [],
  "connectionPoint": [
    {
      "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
      "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
      "name": "TMF635",
      "@referredType": "API"
    }
  ],
  "connectivity": [],
  "activationFeature": [],
  "resourceCharacteristic": []
}
```


Resource Function Specification resource

Specifies a function as a behavior to transform inputs of any nature into outputs of any nature independently from the way it is provided, for example a Medium to Large Enterprise Firewall.

Resource model



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Field descriptions

ResourceFunctionSpecification fields

category	A string. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A string. Description of this REST resource.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
isBundle	A boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).
lastUpdate	A date time (DateTime). Date and time of the last update of this REST resource.
lifecycleStatus	A string. Used to indicate the current lifecycle status of the resource specification.
name	A string. Name given to this REST resource.
version	A string. Resource Specification version.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
connectionPointSpecification	A list of connection point specification references (ConnectionPointSpecificationRef [*]). External connection point specifications. These are the service access points (SAP) where inputs and outputs of the function are available.
connectivitySpecification	A list of resource graph specifications (ResourceGraphSpecification [*]). Internal connectivity potential specifications.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). Complements the description of an element (for instance a resource) through video, pictures ...
featureSpecification	A list of feature specifications (FeatureSpecification [*]). A list of Features for this specification.

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relatedParty	A list of related parties (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
resourceSpecCharacteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic quality or distinctive feature of a ResourceSpecification. The characteristic can be taken on a discrete value, such as color, can take on a range of values, (for example, sensitivity of 100-240 mV), or can be derived from a formula (for example, usage time (hrs) = 30 - talk time *3). Certain characteristics, such as color, may be configured during the ordering or some other process.
resourceSpecRelationship	A list of resource specification relationships (ResourceSpecificationRelationship [*]). A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
targetResourceSchema	A target resource schema (TargetResourceSchema). A target resource schema reference. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A time period. The period for which this REST resource is valid.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.

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content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

CharacteristicValueSpecification sub-resource

specification of a value (number or text or an object) that can be assigned to a Characteristic.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
isDefault	A boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A string. A regular JsonLdExpression constraint for given value.
unitOfMeasure	A string. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A time period. The period of time for which a value is applicable.
value	An any (Any). A discrete value that the characteristic can take on, or the actual value of the characteristic.
valueFrom	An integer. The low range value that a characteristic can take on.
valueTo	An integer. The upper range value that a characteristic can take on.
valueType	A string. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

ConnectionSpecification sub-resource

A specification for an edge in a resource graph.

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href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier for graph edge specification.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
associationType	A string. Association type.
endpointSpecification	A list of endpoint specification references (EndpointSpecificationRef [1..*]). Specifications for resource graph vertices connected by this edge.
name	A string. Descriptive name for graph edge specification.

FeatureSpecification sub-resource

Specification for resource, service or product features.

href	An uri (Uri). Hyperlink reference.
id	A string. Identifier of the feature specification. Must be locally unique within the containing specification, thus allowing direct access to the feature spec.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureSpecCharacteristic	A list of feature specification characteristics (FeatureSpecificationCharacteristic [*]). This is a list of characteristics for a particular feature.
featureSpecRelationship	A list of feature specification relationships (FeatureSpecificationRelationship [*]). A dependency, exclusivity or aggregation relationship between/among feature specifications.
isBundle	A boolean. A flag indicating if this is a feature group (true) or not (false).
isEnabled	A boolean. A flag indicating if the feature is enabled (true) or not (false).
name	A string. Unique name given to the feature specification.
validFor	A time period. The period for which this feature specification is valid.

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version	A string. Version of the feature specification.

FeatureSpecificationCharacteristic sub-resource

Configuration feature characteristic specification.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureSpecCharRelationship	A list of feature specification characteristic relationships (FeatureSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among feature characteristics.
featureSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). Used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a FeatureCharacteristic object.
validFor	A time period. The period of time for which a characteristic is applicable.

FeatureSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among FeatureSpecificationCharacteristics.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicId	A string. Unique identifier of the characteristic within the target feature specification.
featureId	A string. Unique identifier of the target feature specification within the resource specification.
name	A string. Name of the target characteristic.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target feature and feature characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target feature and feature characteristic.
validFor	A time period. The period for which the object is valid.

FeatureSpecificationRelationship sub-resource

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Relationship between feature specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureId	A string. Unique identifier of the target feature specification.
name	A string. This is the name of the target feature specification.
parentSpecificationHref	An uri (Uri). Hyperlink reference to the parent specification containing the target feature.
parentSpecificationId	A string. Unique identifier of the parent specification containing the target feature.
relationshipType	A string. This is the type of the feature specification relationship.
validFor	A time period. The period for which this feature specification relationship is valid.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

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role	A string. Role played by the related party.

ResourceGraphSpecification sub-resource

Resource graph specification.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the resource graph specification.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
connectionSpecification	A list of connection specifications (ConnectionSpecification [1..*]). Resource graph edge specifications.
description	A string. Description of the resource graph specification.
graphSpecificationRelationship	A list of resource graph specification relationships (ResourceGraphSpecificationRelationship [*]). Relationships to other resource graph specifications.
name	A string. Descriptive name for the resource graph specification.

ResourceGraphSpecificationRelationship sub-resource

Describes link between resource graph specifications.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. Semantic of the relationship.
resourceGraph	A resource graph specification reference (ResourceGraphSpecificationRef). A reference to the related resource graph specification.

ResourceSpecificationCharacteristic sub-resource

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This class defines the characteristic features of a resource specification. Every ResourceSpecification has a variety of important attributes, methods, constraints, and
relationships, which distinguish a resource specification from other resource specifications.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A string. When sub-classing, this defines the sub-class Extensible name.
resourceSpecCharRelationship	A list of resource specification characteristic relationships (ResourceSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
resourceSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a ResourceSpecificationCharacteristic object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding ResourceSpecificationCharacteristic object can take on.
validFor	A time period. The period of time for which a characteristic is applicable.

ResourceSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among ResourceSpecificationCharacteristics. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicSpecificationId	A string. Unique identifier of the characteristic within the specification.
name	A string. Name of the target characteristic within the specification.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target characteristic.
validFor	A time period. The period for which the object is valid.

ResourceSpecificationRelationship sub-resource

A migration, substitution, dependency or exclusivity relationship between/among resource specifications.

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@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic that refines the relationship. For example, consider the relationship between a slot and a card. For a half-height card it is important to know the position at which the card is inserted, so a characteristic Position might be defined on the relationship to allow capturing of this in the inventory.
defaultQuantity	An integer. The default number of the related resource that should be instantiated, for example a rack would typically have 4 cards, although it could support more.
href	An uri (Uri). Reference of the target ResourceSpecification.
id	A string. Unique identifier of target ResourceSpecification.
maximumQuantity	An integer. The maximum number of the related resource that should be instantiated, for example a rack supports a maximum of 16 cards.
minimumQuantity	An integer. The minimum number of the related resource that should be instantiated, for example a rack must have at least 1 card.
name	A string. The name given to the target resource specification instance.
relationshipType	A string. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A string. The association role for this resource specification.
validFor	A time period. The period for which the ResourceSpecRelationship is valid.

TargetResourceSchema sub-resource

The reference object to the schema and type of target resource which is described by resource specification.

@schemaLocation	An uri (Uri). This field provides a link to the schema describing the target resource.
@type	A string. Class type of the target resource.

ConnectionPointSpecificationRef relationship

Connection point specification reference. A connection point provides a service access point (SAP) for input and/or output of the resource function.

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@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Connection point specification version.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

EndpointSpecificationRef relationship

A specification for a vertex in a resource graph.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.

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@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
connectionPointSpecification	A connection point specification reference (ConnectionPointSpecificationRef). A specification for the connection point (SAP) of the associated resource function (optional).
isRoot	A boolean. Directionality: true when endpoint is a source, false when a sink. If true for all endpoints connectivity is bidirectional. Default is true.
role	A string. Role of the Resource Function.

ResourceGraphSpecificationRef relationship

Resource graph specification reference.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

Provided below is the json representation of an example of a 'ResourceFunctionSpecification' resource object


```
{
  "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
  "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
  "name": "Encabulator Function Specification",
  "description": "This specification describes the Encabulator Function",
  "@type": "ResourceFunctionSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lastUpdate": "2022-02-03T07:59:59Z",
  "lifecycleStatus": "Active",
```


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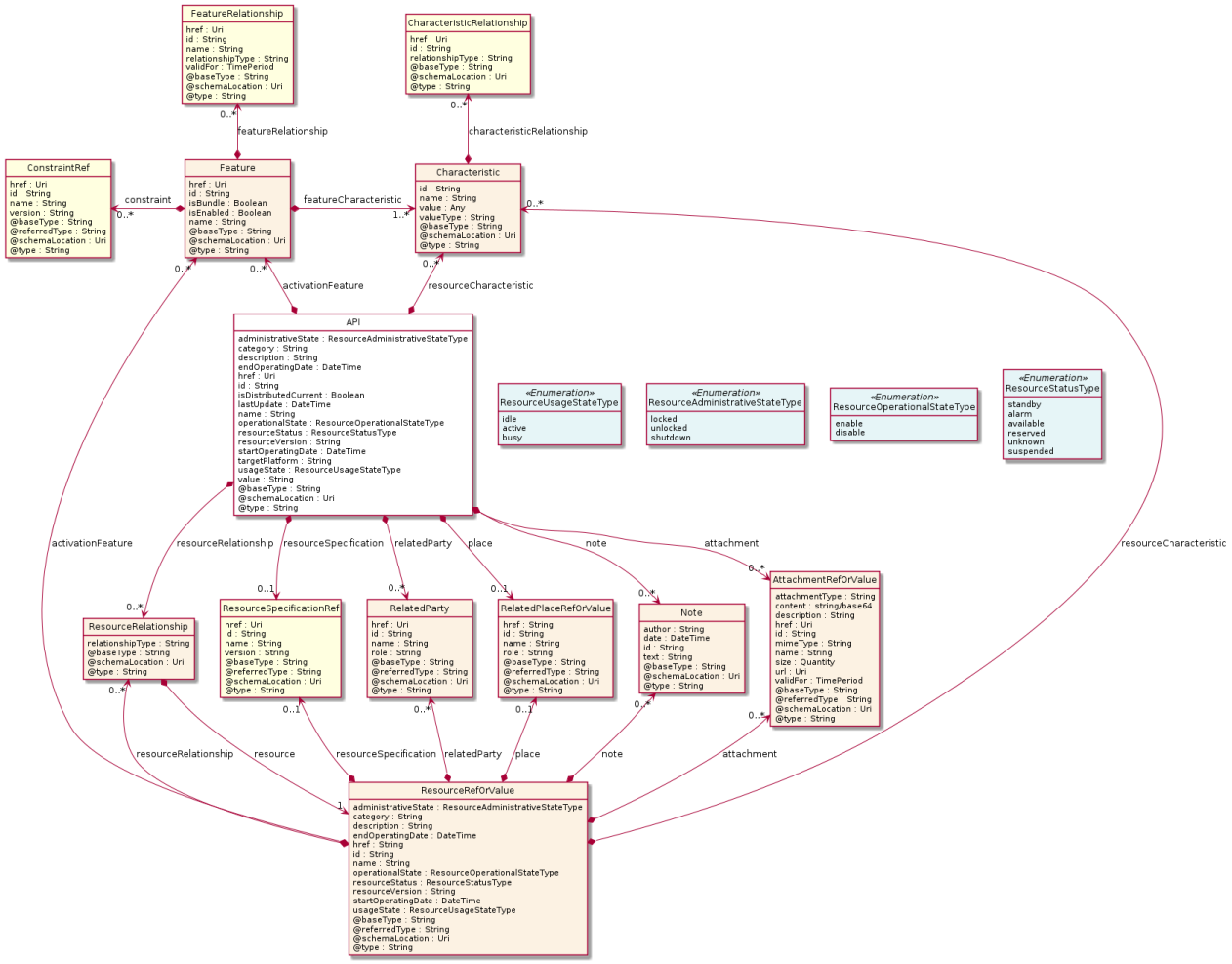
```
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "ResourceFunction"
  },
  "resourceSpecRelationship": [],
  "featureSpecification": [],
  "resourceSpecCharacteristic": [
    {
      "name": "numMarzelvanes",
      "description": "Number of antigravic marzelvanes.",
      "valueType": "integer",
      "resourceSpecCharacteristicValue": [
        {
          "valueFrom": 6,
          "valueTo": 32
        }
      ]
    },
    {
      "name": "numStatorSlots",
      "description": "Number of panandermic semi-boloid stator slots.",
      "valueType": "integer",
      "resourceSpecCharacteristicValue": [
        {
          "valueFrom": 0,
          "valueTo": 50
        }
      ]
    }
  ],
  "connectionPointSpecification": [
    {
      "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
      "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
      "name": "TMF635 Specification",
      "@referredType": "APISpecification"
    }
  ],
  "connectivitySpecification": []
}
```


A P I resource

An Application Program Interface (API) is a set of routines, protocols, and tools for building software applications.

Resource model

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Field descriptions

API fields

isDistributedCurrent	A boolean. True defines this SoftwareResource as being currently distributed across several Resource nodes.
lastUpdate	A date time (DateTime). The time of the last update of this resource.
targetPlatform	A string. Defines the target platform that the SoftwareResource will be deployed on.
value	A string. the value of the logical resource. E.g. '0746712345' for MSISDN's.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.
endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.

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href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
name	A string. A string used to give a name to the resource.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
activationFeature	A list of features (Feature [*]). Configuration features.
administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.
note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource

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	needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

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attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

Feature sub-resource

Configuration feature.

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href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureCharacteristic	A list of characteristics (Characteristic [1..*]). This is a list of Characteristics for a particular feature.
featureRelationship	A list of feature relationships (FeatureRelationship [*]). Configuration feature.
isBundle	A boolean. True if this is a feature group. Default is false.
isEnabled	A boolean. True if this feature is enabled. Default is true.
name	A string. This is the name for the feature.

FeatureRelationship sub-resource

Configuration feature.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the target feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
name	A string. This is the name of the target feature.
relationshipType	A string. This is the type of the feature relationship.
validFor	A time period. The period for which this feature relationship is valid.

Note sub-resource

Extra information about a given entity.

@baseType	A string. When sub-classing, this defines the super-class.

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@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.
text	A string. Text of the note.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

RelatedPlaceRefOrValue sub-resource

Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.

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name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

ResourceRefOrValue sub-resource

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the related ResourceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A string used to give a name to the resource.
href	A string. The URI for the object itself.
id	A string. Identifier of an instance of the resource. Required to be unique within the resource type. Used in URIs as the identifier for specific instances of a type.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.
endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.

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activationFeature	A list of features (Feature [*]). Configuration features.
administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.
note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

ResourceRelationship sub-resource

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g.: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A resource ref or value (ResourceRefOrValue). Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the related ResourceRefOrValue class itself.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.

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@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

ResourceSpecificationRef relationship

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components.
A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Resource Specification version.

Json representation sample

Provided below is the json representation of an example of a 'API' resource object


```
{
  "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
  "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
  "name": "TMF635",
  "description": "Open API for Usage Management",
  "@type": "API",
  "startOperatingDate": "2021-07-01T08:00:00Z",
  "lastUpdate": "2022-11-27T20:51:14.457Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
    "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
```

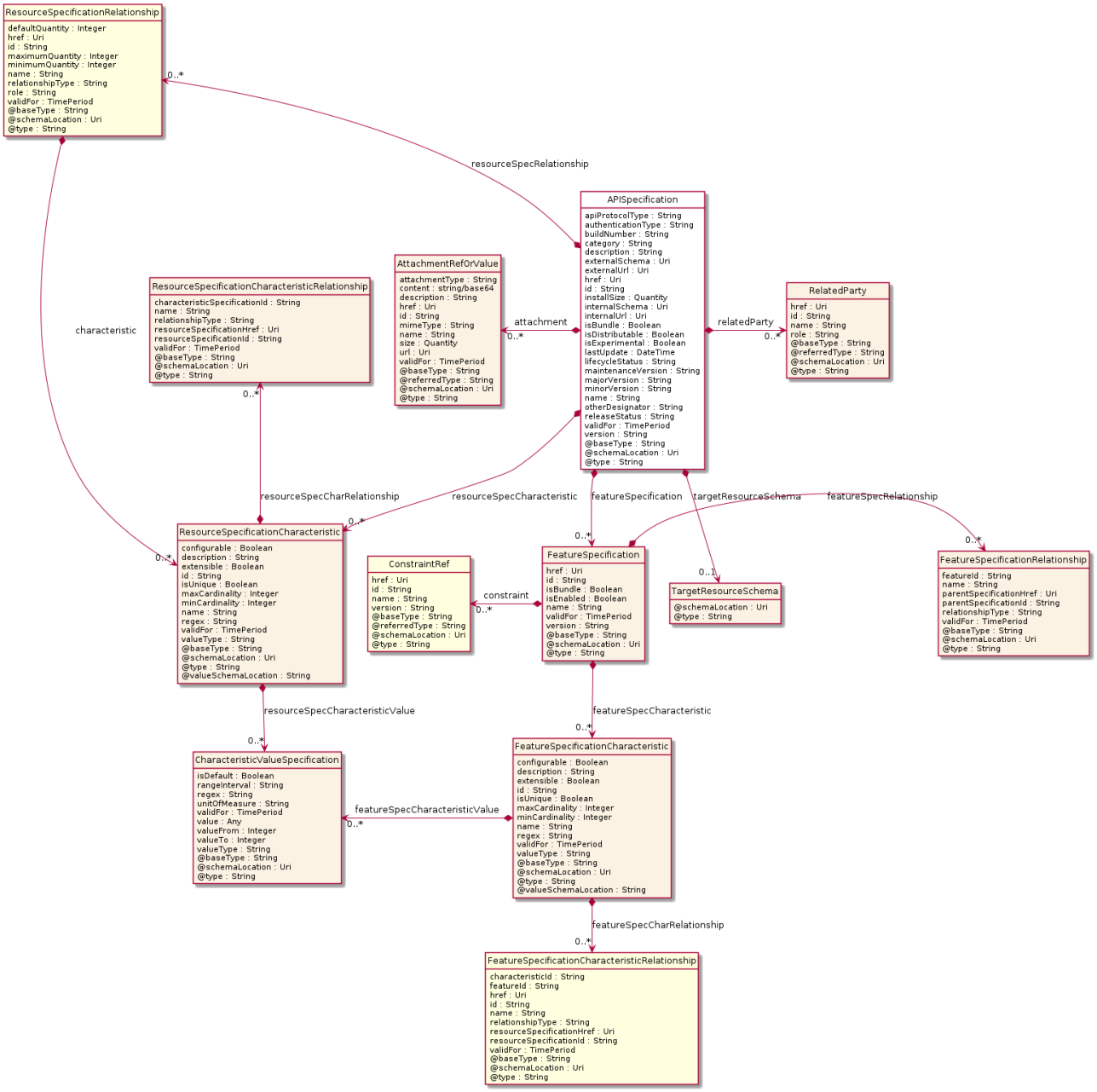

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```
      "name": "TMF635 Specification",
      "@referredType": "APISpecification"
    },
    "resourceRelationship": [],
    "resourceCharacteristic": []
}
```

A P I Specification resource

A base class is used to define the invariant characteristics and behavior (attributes, constraints, and relationships) of an API.

Resource model



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Field descriptions

APISpecification fields

buildNumber	A string. Defines the internal identifier for the compilation of this unit of a SoftwareResourceSpecification.
isDistributable	A boolean. Designates the SoftwareResource as being able to be distributed across multiple physically distinct Resources.
isExperimental	A boolean. True represents a special build (perhaps as a trial to suit specific customer needs).
maintenanceVersion	A string. Defines the maintenance number component of the version of SoftwareResourceSpecification.
majorVersion	A string. Defines the major number component of the version of SoftwareResourceSpecification.
minorVersion	A string. Defines the minor number component of the version of SoftwareResourceSpecification.
otherDesignator	A string. Enables the manufacturer to label this software as having some set of special restrictions.
releaseStatus	A string. Defines the particular release status of this SoftwareResourceSpecification.
category	A string. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A string. Description of this REST resource.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
isBundle	A boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).
lastUpdate	A date time (DateTime). Date and time of the last update of this REST resource.
lifecycleStatus	A string. Used to indicate the current lifecycle status of the resource specification.
name	A string. Name given to this REST resource.
version	A string. Resource Specification version.
@baseType	A string. When sub-classing, this defines the super-class.

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@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
apiProtocolType	A string. API protocol type like REST, SOAP and so on.
authenticationType	A string. The authentication scheme used for this API.
externalSchema	An uri (Uri). The external API schema of the (external) URL offered to the client of this API.
externalUrl	An uri (Uri). The URL of an API offered to the user. This is seen as external url for an API brokerage.
internalSchema	An uri (Uri). The internal API schema of the (internal) URL which an API broker uses.
internalUrl	An uri (Uri). The URL of an API offered to an API brokerage.
installSize	A quantity (Quantity). The amount and units (e.g. 800MB) required to install the SoftwareResource.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). Complements the description of an element (for instance a resource) through video, pictures ...
featureSpecification	A list of feature specifications (FeatureSpecification [*]). A list of Features for this specification.
relatedParty	A list of related parties (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
resourceSpecCharacteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic quality or distinctive feature of a ResourceSpecification. The characteristic can be taken on a discrete value, such as color, can take on a range of values, (for example, sensitivity of 100-240 mV), or can be derived from a formula (for example, usage time (hrs) = 30 - talk time *3). Certain characteristics, such as color, may be configured during the ordering or some other process.
resourceSpecRelationship	A list of resource specification relationships (ResourceSpecificationRelationship [*]). A migration, substitution, dependency or exclusivity relationship between/among resource specifications.

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targetResourceSchema	A target resource schema (TargetResourceSchema). A target resource schema reference. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A time period. The period for which this REST resource is valid.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

CharacteristicValueSpecification sub-resource

specification of a value (number or text or an object) that can be assigned to a Characteristic.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A string. When sub-classing, this defines the sub-class Extensible name.
isDefault	A boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A string. A regular JsonLdExpression constraint for given value.
unitOfMeasure	A string. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A time period. The period of time for which a value is applicable.
value	An any (Any). A discrete value that the characteristic can take on, or the actual value of the characteristic.
valueFrom	An integer. The low range value that a characteristic can take on.
valueTo	An integer. The upper range value that a characteristic can take on.
valueType	A string. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

FeatureSpecification sub-resource

Specification for resource, service or product features.

href	An uri (Uri). Hyperlink reference.
id	A string. Identifier of the feature specification. Must be locally unique within the containing specification, thus allowing direct access to the feature spec.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureSpecCharacteristic	A list of feature specification characteristics (FeatureSpecificationCharacteristic [*]). This is a list of characteristics for a particular feature.

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featureSpecRelationship	A list of feature specification relationships (FeatureSpecificationRelationship [*]). A dependency, exclusivity or aggregation relationship between/among feature specifications.
isBundle	A boolean. A flag indicating if this is a feature group (true) or not (false).
isEnabled	A boolean. A flag indicating if the feature is enabled (true) or not (false).
name	A string. Unique name given to the feature specification.
validFor	A time period. The period for which this feature specification is valid.
version	A string. Version of the feature specification.

FeatureSpecificationCharacteristic sub-resource

Configuration feature characteristic specification.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.

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regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureSpecCharRelationship	A list of feature specification characteristic relationships (FeatureSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among feature characteristics.
featureSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). Used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a FeatureCharacteristic object.
validFor	A time period. The period of time for which a characteristic is applicable.

FeatureSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among FeatureSpecificationCharacteristics.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicId	A string. Unique identifier of the characteristic within the target feature specification.
featureId	A string. Unique identifier of the target feature specification within the resource specification.
name	A string. Name of the target characteristic.

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relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target feature and feature characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target feature and feature characteristic.
validFor	A time period. The period for which the object is valid.

FeatureSpecificationRelationship sub-resource

Relationship between feature specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureId	A string. Unique identifier of the target feature specification.
name	A string. This is the name of the target feature specification.
parentSpecificationHref	An uri (Uri). Hyperlink reference to the parent specification containing the target feature.
parentSpecificationId	A string. Unique identifier of the parent specification containing the target feature.
relationshipType	A string. This is the type of the feature specification relationship.
validFor	A time period. The period for which this feature specification relationship is valid.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.

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name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

ResourceSpecificationCharacteristic sub-resource

This class defines the characteristic features of a resource specification. Every ResourceSpecification has a variety of important attributes, methods, constraints, and
relationships, which distinguish a resource specification from other resource specifications.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone

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	numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
resourceSpecCharRelationship	A list of resource specification characteristic relationships (ResourceSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
resourceSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a ResourceSpecificationCharacteristic object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding ResourceSpecificationCharacteristic object can take on.
validFor	A time period. The period of time for which a characteristic is applicable.

ResourceSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among ResourceSpecificationCharacteristics. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

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characteristicSpecificationId	A string. Unique identifier of the characteristic within the specification.
name	A string. Name of the target characteristic within the specification.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target characteristic.
validFor	A time period. The period for which the object is valid.

ResourceSpecificationRelationship sub-resource

A migration, substitution, dependency or exclusivity relationship between/among resource specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic that refines the relationship. For example, consider the relationship between a slot and a card. For a half-height card it is important to know the position at which the card is inserted, so a characteristic Position might be defined on the relationship to allow capturing of this in the inventory.
defaultQuantity	An integer. The default number of the related resource that should be instantiated, for example a rack would typically have 4 cards, although it could support more.
href	An uri (Uri). Reference of the target ResourceSpecification.
id	A string. Unique identifier of target ResourceSpecification.
maximumQuantity	An integer. The maximum number of the related resource that should be instantiated, for example a rack supports a maximum of 16 cards.
minimumQuantity	An integer. The minimum number of the related resource that should be instantiated, for example a rack must have at least 1 card.
name	A string. The name given to the target resource specification instance.
relationshipType	A string. Type of relationship such as migration, substitution, dependency, exclusivity.

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role	A string. The association role for this resource specification.
validFor	A time period. The period for which the ResourceSpecRelationship is valid.

TargetResourceSchema sub-resource

The reference object to the schema and type of target resource which is described by resource specification.

@schemaLocation	An uri (Uri). This field provides a link to the schema describing the target resource.
@type	A string. Class type of the target resource.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

Json representation sample

Provided below is the json representation of an example of a 'APISpecification' resource object


```
{
  "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
  "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
  "name": "TMF635 Specification",
  "description": "Open API for Usage Management",
  "@type": "APISpecification",
  "validFor": {
    "startDateTime": "2021-07-01T08:00:00Z"
  },
  "lifecycleStatus": "Active",
  "lastUpdate": "2022-11-27T07:59:59Z",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
```


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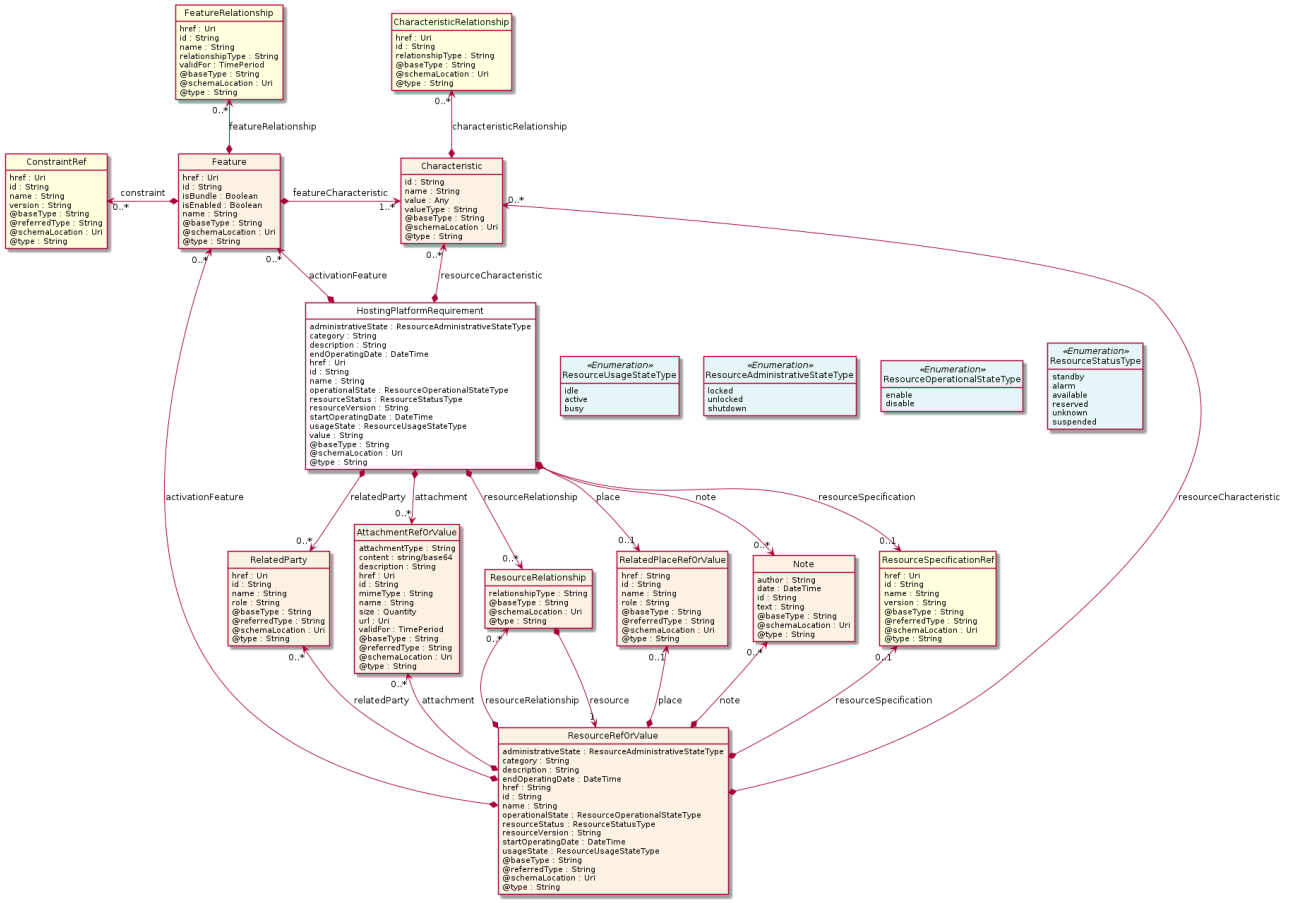
```
      "@type": "API"
    },
    "releaseStatus": "generalDeployment",
    "buildNumber": "20.5.0",
    "majorVersion": "4",
    "minorVersion": "0",
    "maintenanceVersion": "0",
    "isDistributable": true,
    "isExperimental": false,
    "apiProtocolType": "REST",
    "externalUrl": "https://example.net/tmf-api/usageManagement/v4",
    "internalUrl": "/tmf-api/usageManagement/v4",
    "externalSchema": "https://tmf-open-api-table-documents.s3.eu-west-
1.amazonaws.com/OpenAPITable/4.0.0/swagger/TMF635-UsageManagement-v4.0.0.swagger.json",
    "internalSchema": "/tmf-api/usageManagement/v4/schema/TMF635-UsageManagement-v4.0.0.swagger.json",
    "authenticationType": "Basic",
    "allowedOperations": [
      "GET",
      "HEAD",
      "POST",
      "PATCH",
      "DELETE"
    ],
    "allowedAPIEntities": [
      "usageSpecification",
      "usage"
    ],
    "responseTypeFormat": [
      "application/json",
      "application/problem+json"
    ],
    "resourceSpecRelationship": [],
    "relatedParty": [
      {
        "id": "ab5660aa-064d-440c-b7d1-b943e42b904e",
        "href": "/partyManagement/v4/organization/ab5660aa-064d-440c-b7d1-b943e42b904e",
        "name": "TM Forum",
        "role": "supplier"
      }
    ],
    "resourceSpecCharacteristic": []
}
```


Hosting Platform Requirement resource

A HostingPlatformRequirement implements a HostingPlatformRequirementSpecification for a specific InstalledSoftware.

Resource model

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Field descriptions

HostingPlatformRequirement fields

value	A string. the value of the logical resource. E.g. '0746712345' for MSISDN's.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.
endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
name	A string. A string used to give a name to the resource.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.

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href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
activationFeature	A list of features (Feature [*]). Configuration features.
administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.
note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the

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	Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.

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validFor	A time period. The period of time for which the attachment is valid.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

Feature sub-resource

Configuration feature.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureCharacteristic	A list of characteristics (Characteristic [1..*]). This is a list of Characteristics for a particular feature.
featureRelationship	A list of feature relationships (FeatureRelationship [*]). Configuration feature.
isBundle	A boolean. True if this is a feature group. Default is false.
isEnabled	A boolean. True if this feature is enabled. Default is true.
name	A string. This is the name for the feature.

FeatureRelationship sub-resource

Configuration feature.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the target feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
name	A string. This is the name of the target feature.
relationshipType	A string. This is the type of the feature relationship.
validFor	A time period. The period for which this feature relationship is valid.

Note sub-resource

Extra information about a given entity.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.

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text	A string. Text of the note.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

RelatedPlaceRefOrValue sub-resource

Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

ResourceRefOrValue sub-resource

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the related ResourceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A string used to give a name to the resource.
href	A string. The URI for the object itself.
id	A string. Identifier of an instance of the resource. Required to be unique within the resource type. Used in URIs as the identifier for specific instances of a type.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.
endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.
activationFeature	A list of features (Feature [*]). Configuration features.
administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

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note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

ResourceRelationship sub-resource

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned

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resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g.: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A resource ref or value (ResourceRefOrValue). Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the related ResourceRefOrValue class itself.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

ResourceSpecificationRef relationship

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network

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Elements, software, IT systems, content and information, and technology components.
A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Resource Specification version.

Json representation sample

Provided below is the json representation of an example of a 'HostingPlatformRequirement' resource object


```
{
  "id": "a1976344-3933-4942-874a-ded80c1e5d83",
  "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
  "name": "Encabulator Platform Requirement",
  "description": "The requirements on platforms hosting the Encabulator software",
  "@type": "HostingPlatformRequirement",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Compute",
  "resourceSpecification": {
    "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
    "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
    "name": "Encabulator Platform Requirement Specification",
    "@referredType": "HostingPlatformRequirementSpecification"
  },
  "resourceRelationship": [],
  "activationFeature": [
    {
      "name": "scale",
      "featureCharacteristic": [
        {
          "name": "cpuSize",
          "value": "micro"
        },
```


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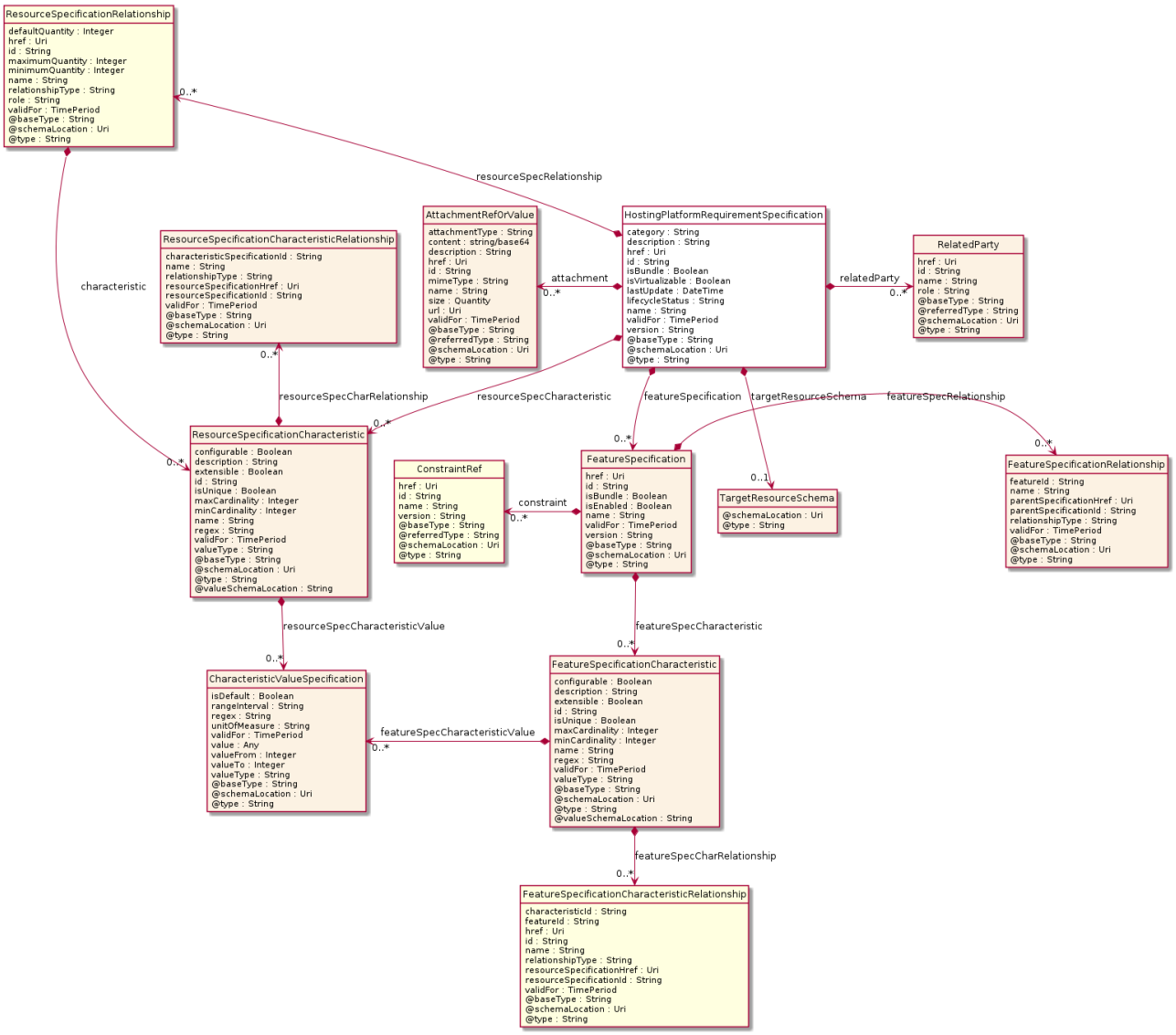
```
            {
              "name": "cpuNum",
              "value": 2
            },
            {
              "name": "memory",
              "value": 1024
            }
          ]
        }
      ],
      "resourceCharacteristic": [
        {
          "name": "virtualization",
          "value": "container-runtime"
        }
      ]
}
```


Hosting Platform Requirement Specification resource

This is a base class that is used to define the invariant characteristics and behavior (attributes, constraints, and relationships) of a HostingPlatformRequirement Resource.

Resource model

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Field descriptions

HostingPlatformRequirementSpecification fields

category	A string. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A string. Description of this REST resource.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
isBundle	A boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).

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lastUpdate	A date time (DateTime). Date and time of the last update of this REST resource.
lifecycleStatus	A string. Used to indicate the current lifecycle status of the resource specification.
name	A string. Name given to this REST resource.
version	A string. Resource Specification version.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
isVirtualizable	A boolean. True if the Capacity required can be virtualized.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). Complements the description of an element (for instance a resource) through video, pictures ...
featureSpecification	A list of feature specifications (FeatureSpecification [*]). A list of Features for this specification.
relatedParty	A list of related parties (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
resourceSpecCharacteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic quality or distinctive feature of a ResourceSpecification. The characteristic can be taken on a discrete value, such as color, can take on a range of values, (for example, sensitivity of 100-240 mV), or can be derived from a formula (for example, usage time (hrs) = 30 - talk time *3). Certain characteristics, such as color, may be configured during the ordering or some other process.
resourceSpecRelationship	A list of resource specification relationships (ResourceSpecificationRelationship [*]). A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
targetResourceSchema	A target resource schema (TargetResourceSchema). A target resource schema reference. The reference object to the schema and type of target resource which is described by resource specification.

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validFor	A time period. The period for which this REST resource is valid.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

CharacteristicValueSpecification sub-resource

specification of a value (number or text or an object) that can be assigned to a Characteristic.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
isDefault	A boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.

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rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A string. A regular JsonLdExpression constraint for given value.
unitOfMeasure	A string. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A time period. The period of time for which a value is applicable.
value	An any (Any). A discrete value that the characteristic can take on, or the actual value of the characteristic.
valueFrom	An integer. The low range value that a characteristic can take on.
valueTo	An integer. The upper range value that a characteristic can take on.
valueType	A string. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

FeatureSpecification sub-resource

Specification for resource, service or product features.

href	An uri (Uri). Hyperlink reference.
id	A string. Identifier of the feature specification. Must be locally unique within the containing specification, thus allowing direct access to the feature spec.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureSpecCharacteristic	A list of feature specification characteristics (FeatureSpecificationCharacteristic [*]). This is a list of characteristics for a particular feature.
featureSpecRelationship	A list of feature specification relationships (FeatureSpecificationRelationship [*]). A dependency, exclusivity or aggregation relationship between/among feature specifications.

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isBundle	A boolean. A flag indicating if this is a feature group (true) or not (false).
isEnabled	A boolean. A flag indicating if the feature is enabled (true) or not (false).
name	A string. Unique name given to the feature specification.
validFor	A time period. The period for which this feature specification is valid.
version	A string. Version of the feature specification.

FeatureSpecificationCharacteristic sub-resource

Configuration feature characteristic specification.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.

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valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureSpecCharRelationship	A list of feature specification characteristic relationships (FeatureSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among feature characteristics.
featureSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). Used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a FeatureCharacteristic object.
validFor	A time period. The period of time for which a characteristic is applicable.

FeatureSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among FeatureSpecificationCharacteristics.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicId	A string. Unique identifier of the characteristic within the target feature specification.
featureId	A string. Unique identifier of the target feature specification within the resource specification.
name	A string. Name of the target characteristic.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.

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resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target feature and feature characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target feature and feature characteristic.
validFor	A time period. The period for which the object is valid.

FeatureSpecificationRelationship sub-resource

Relationship between feature specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureId	A string. Unique identifier of the target feature specification.
name	A string. This is the name of the target feature specification.
parentSpecificationHref	An uri (Uri). Hyperlink reference to the parent specification containing the target feature.
parentSpecificationId	A string. Unique identifier of the parent specification containing the target feature.
relationshipType	A string. This is the type of the feature specification relationship.
validFor	A time period. The period for which this feature specification relationship is valid.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.

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id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

ResourceSpecificationCharacteristic sub-resource

This class defines the characteristic features of a resource specification. Every ResourceSpecification has a variety of important attributes, methods, constraints, and
relationships, which distinguish a resource specification from other resource specifications.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.

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name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
resourceSpecCharRelationship	A list of resource specification characteristic relationships (ResourceSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
resourceSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a ResourceSpecificationCharacteristic object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding ResourceSpecificationCharacteristic object can take on.
validFor	A time period. The period of time for which a characteristic is applicable.

ResourceSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among ResourceSpecificationCharacteristics. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicSpecificationId	A string. Unique identifier of the characteristic within the specification.

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name	A string. Name of the target characteristic within the specification.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target characteristic.
validFor	A time period. The period for which the object is valid.

ResourceSpecificationRelationship sub-resource

A migration, substitution, dependency or exclusivity relationship between/among resource specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic that refines the relationship. For example, consider the relationship between a slot and a card. For a half-height card it is important to know the position at which the card is inserted, so a characteristic Position might be defined on the relationship to allow capturing of this in the inventory.
defaultQuantity	An integer. The default number of the related resource that should be instantiated, for example a rack would typically have 4 cards, although it could support more.
href	An uri (Uri). Reference of the target ResourceSpecification.
id	A string. Unique identifier of target ResourceSpecification.
maximumQuantity	An integer. The maximum number of the related resource that should be instantiated, for example a rack supports a maximum of 16 cards.
minimumQuantity	An integer. The minimum number of the related resource that should be instantiated, for example a rack must have at least 1 card.
name	A string. The name given to the target resource specification instance.
relationshipType	A string. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A string. The association role for this resource specification.

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validFor	A time period. The period for which the ResourceSpecRelationship is valid.

TargetResourceSchema sub-resource

The reference object to the schema and type of target resource which is described by resource specification.

@schemaLocation	An uri (Uri). This field provides a link to the schema describing the target resource.
@type	A string. Class type of the target resource.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

Json representation sample

Provided below is the json representation of an example of a 'HostingPlatformRequirementSpecification' resource object


```
{
  "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
  "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
  "name": "Encabulator Platform Requirement Specification",
  "description": "This specification describes requirements on platforms hosting the Encabulator software",
  "@type": "HostingPlatformRequirementSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Compute",
  "targetResourceSchema": {
    "@type": "HostingPlatformRequirement"
```


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```
  },
  "resourceSpecRelationship": [],
  "featureSpecification": [
    {
      "name": "scale",
      "featureSpecCharacteristic": [
        {
          "name": "cpuSize",
          "description": "Vertical CPU scaling dimension",
          "valueType": "string",
          "featureSpecCharacteristicValue": [
            {
              "isDefault": true,
              "value": "micro"
            },
            {
              "isDefault": false,
              "value": "small"
            },
            {
              "isDefault": false,
              "value": "medium"
            },
            {
              "isDefault": false,
              "value": "large"
            }
          ]
        },
        {
          "name": "cpuNum",
          "description": "Minimum number of CPUs",
          "valueType": "integer",
          "featureSpecCharacteristicValue": [
            {
              "valueFrom": 1,
              "valueTo": 24
            }
          ]
        },
        {
          "name": "memory",
          "description": "Minimum amount of memory required",
          "valueType": "integer",
          "featureSpecCharacteristicValue": [
            {
              "unitOfMeasure": "MiB",
              "valueFrom": 640,
              "valueTo": 16384
            }
          ]
        }
      ]
    }
  ],
```


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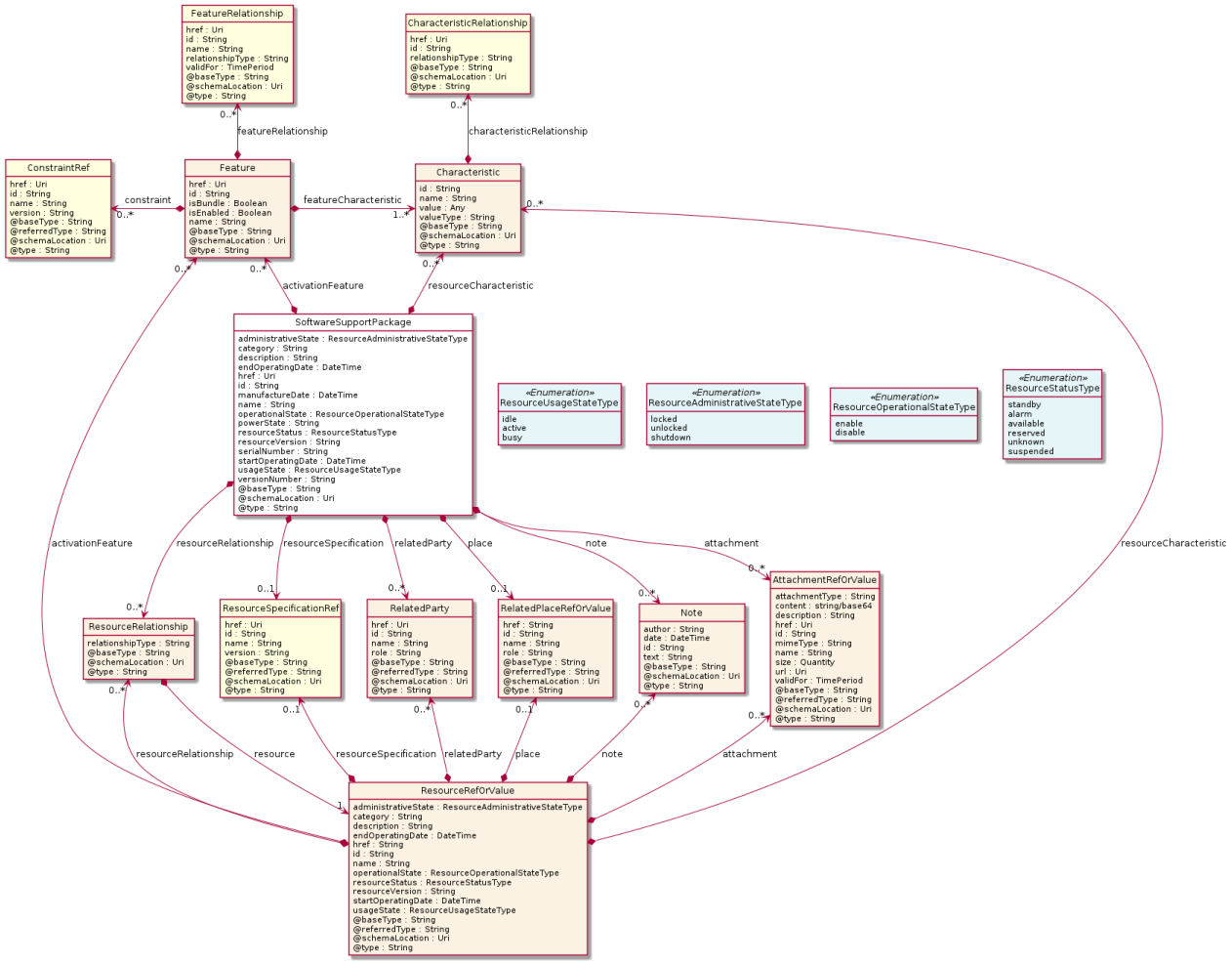
```
    "resourceSpecCharacteristic": [
      {
        "name": "virtualization",
        "description": "Type of virtualization required",
        "valueType": "string",
        "resourceSpecCharacteristicValue": [
          {
            "isDefault": true,
            "value": "container-runtime"
          },
          {
            "isDefault": false,
            "value": "hypervisor"
          }
        ]
      }
    ]
}
```

Software Support Package resource

A SofwareSupportPackage represents the package acquired by a consumer from a software vendor. It can be materialized as one or several files (data) downloaded online or copied on a physical support. It contains all files required for deployment and installation such as installation manager and at minimum, one of the files constitutes the set of machine readable instructions that will be executed automatically after it is installed.

Resource model

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Field descriptions

SoftwareSupportPackage fields

manufactureDate	A date time (DateTime). This is a string attribute that defines the date of manufacture of this item in the fixed format "dd/mm/yyyy". This is an optional attribute.
powerState	A string. This defines the current power status of the hardware item. Values include: 0: Unknown 1: Not Applicable 2: No Power Applied 3: Full Power Applied 4: Power Save - Normal 5: Power Save - Degraded 6: Power Save - Standby 7: Power Save - Critical 8: Power Save - Low Power Mode 9: Power Save - Unknown

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	10: Power Cycle 11: Power Warning 12: Power Off.
serialNumber	A string. This is a string that represents a manufacturer-allocated number used to identify different instances of the same hardware item. The ModelNumber and PartNumber attributes are used to identify different types of hardware items. This is a REQUIRED attribute.
versionNumber	A string. This is a string that identifies the version of this physical resource. This is an optional attribute.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.
endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
name	A string. A string used to give a name to the resource.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
activationFeature	A list of features (Feature [*]). Configuration features.
administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

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note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

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@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.

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valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

Feature sub-resource

Configuration feature.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureCharacteristic	A list of characteristics (Characteristic [1..*]). This is a list of Characteristics for a particular feature.
featureRelationship	A list of feature relationships (FeatureRelationship [*]). Configuration feature.
isBundle	A boolean. True if this is a feature group. Default is false.
isEnabled	A boolean. True if this feature is enabled. Default is true.
name	A string. This is the name for the feature.

FeatureRelationship sub-resource

Configuration feature.

href	An uri (Uri). Hyperlink reference.

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id	A string. Unique identifier of the target feature.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
name	A string. This is the name of the target feature.
relationshipType	A string. This is the type of the feature relationship.
validFor	A time period. The period for which this feature relationship is valid.

Note sub-resource

Extra information about a given entity.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.
text	A string. Text of the note.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.

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id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

RelatedPlaceRefOrValue sub-resource

Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

ResourceRefOrValue sub-resource

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the related ResourceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A string used to give a name to the resource.
href	A string. The URI for the object itself.
id	A string. Identifier of an instance of the resource. Required to be unique within the resource type. Used in URIs as the identifier for specific instances of a type.

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@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A string. Category of the concrete resource. e.g. Gold, Silver for MSISDN concrete resource.
description	A string. free-text description of the resource.
endOperatingDate	A date time (DateTime). A date time(DateTime). The date till the resource is operating.
resourceVersion	A string. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A date time (DateTime). A date time(DateTime). The date from which the resource is operating.
activationFeature	A list of features (Feature [*]). Configuration features.
administrativeState	A resource administrative state type (ResourceAdministrativeStateType). Tracks the administrative state of the resource, such as locked, unlocked, shutdown and so on.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.
note	A list of notes (Note [*]). Extra information about a given entity.
operationalState	A resource operational state type (ResourceOperationalStateType). Tracks the operational state of the resource, such as enable, disable and so on.
place	A related place ref or value (RelatedPlaceRefOrValue). Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.
relatedParty	A list of related parties (RelatedParty [*]). Related Entity reference. A related party defines party or party role linked to a specific entity.
resourceCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
resourceRelationship	A list of resource relationships (ResourceRelationship [*]). Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already

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	owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A resource specification reference (ResourceSpecificationRef). Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A resource status type (ResourceStatusType). Tracks the resource status of the resource, such as standby, alarm, available, reserved, suspended and so on.
usageState	A resource usage state type (ResourceUsageStateType). Tracks the usage state of the resource, such as idle, active, busy and so on.

ResourceRelationship sub-resource

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g.: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A resource ref or value (ResourceRefOrValue). Resource is an abstract entity that describes the common set of attributes shared by all concrete resources. The polymorphic attributes @type, @schemaLocation & @referredType are

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related to the Resource entity and not the related ResourceRefOrValue class itself.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

ResourceSpecificationRef relationship

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components.
A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Resource Specification version.

Json representation sample

Provided below is the json representation of an example of a 'SoftwareSupportPackage' resource object

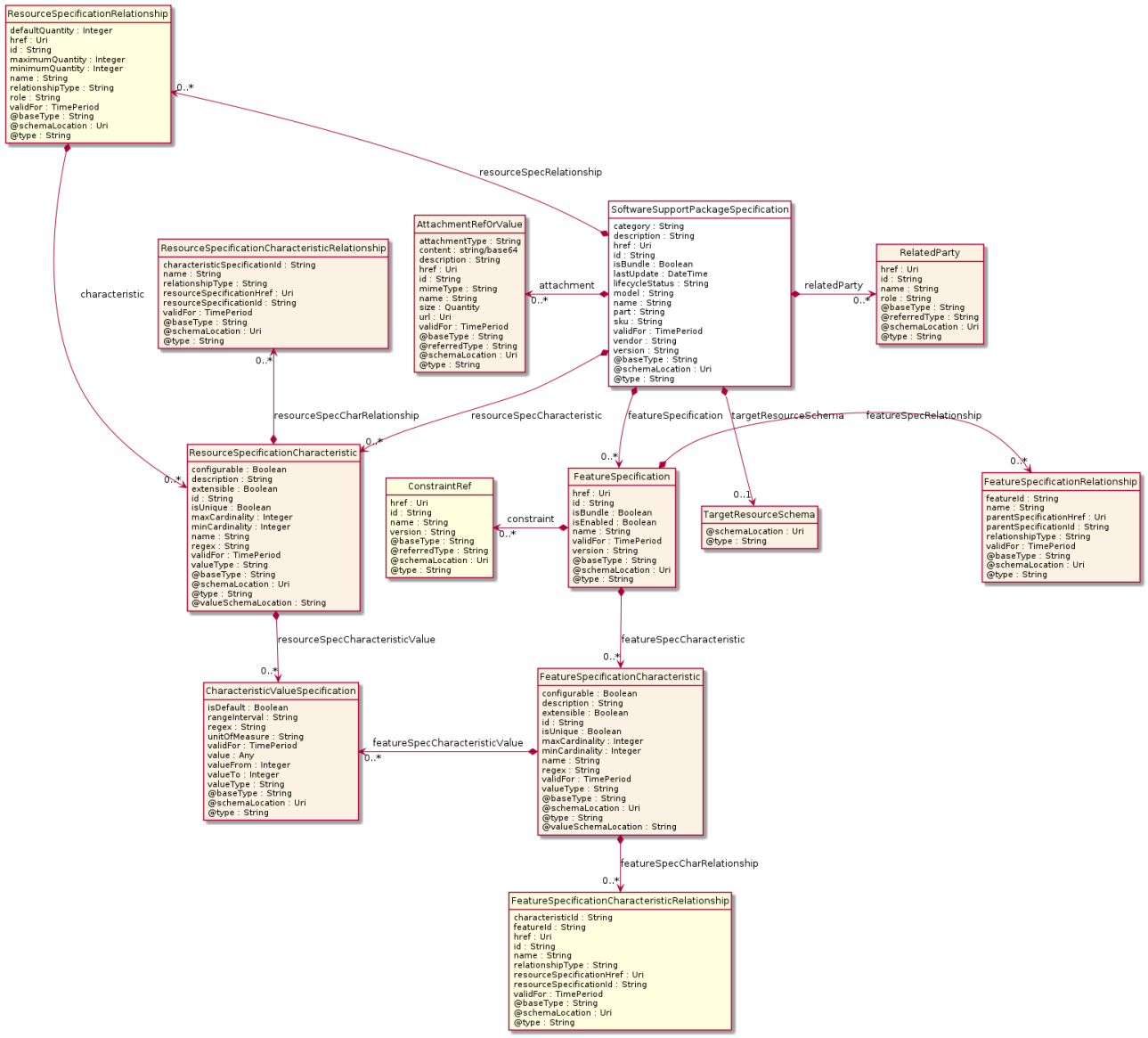

```
{
  "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
  "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
  "name": "Encabulator Software Package",
  "description": "Package providing the files containing the Encabulator software",
  "@type": "SoftwareSupportPackage",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "632547af-e850-45a1-a569-31531ea81eb7",
    "href": "/resourceCatalog/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7",
    "name": "Software Package Repository Specification"
  },
  "resourceRelationship": [],
  "resourceCharacteristic": [
    {
      "name": "repoName",
      "value": "Local"
    },
    {
      "name": "repoBase",
      "value": "https://repo.example.net"
    },
    {
      "name": "imagePath",
      "value": "/cnf/9c53e70b-5e78-41e8-826b-8c70c2b65a8e"
    }
  ]
}
```


Software Support Package Specification resource

A base class is used to define the invariant characteristics and behavior (attributes, constraints, and relationships) of a SoftwareSupportPackage.

Resource model

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Field descriptions

SoftwareSupportPackageSpecification fields

model	A string. This is a string that represents a manufacturer-allocated number used to identify the general type and/or category of the hardware item. This, in combination with the Part and the Vendor, identify different types of hardware items. The SerialNumber can then be used to differentiate between different instances of the same type of hardware item. This is an optional attribute.
part	A string. This is a string that defines a manufacturer-allocated part number assigned by the organization that manufactures the hardware item. This, in combination with the Model and the Vendor, identify different types of hardware items. The SerialNumber can then be used to

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	differentiate between different instances of the same type of hardware item. This is a REQUIRED attribute.
sku	A string. This is a string that defines the manufacturer-allocated Stock Keeping Unit (SKU) number of the hardware item. This is an optional attribute.
vendor	A string. This is a string that defines the name of the manufacturer. This, in combination with the Model and the Part, identify different types of hardware items. The SerialNumber can then be used to differentiate between different instances of the same type of hardware item. This is a REQUIRED attribute for a physical resource.
category	A string. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A string. Description of this REST resource.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
isBundle	A boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).
lastUpdate	A date time (DateTime). Date and time of the last update of this REST resource.
lifecycleStatus	A string. Used to indicate the current lifecycle status of the resource specification.
name	A string. Name given to this REST resource.
version	A string. Resource Specification version.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). Complements the description of an element (for instance a resource) through video, pictures ...

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featureSpecification	A list of feature specifications (FeatureSpecification [*]). A list of Features for this specification.
relatedParty	A list of related parties (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
resourceSpecCharacteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic quality or distinctive feature of a ResourceSpecification. The characteristic can be taken on a discrete value, such as color, can take on a range of values, (for example, sensitivity of 100-240 mV), or can be derived from a formula (for example, usage time (hrs) = 30 - talk time *3). Certain characteristics, such as color, may be configured during the ordering or some other process.
resourceSpecRelationship	A list of resource specification relationships (ResourceSpecificationRelationship [*]). A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
targetResourceSchema	A target resource schema (TargetResourceSchema). A target resource schema reference. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A time period. The period for which this REST resource is valid.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Remote reference to the content if web-addressable.
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

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attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

CharacteristicValueSpecification sub-resource

specification of a value (number or text or an object) that can be assigned to a Characteristic.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
isDefault	A boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A string. A regular JsonLdExpression constraint for given value.
unitOfMeasure	A string. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A time period. The period of time for which a value is applicable.
value	An any (Any). A discrete value that the characteristic can take on, or the actual value of the characteristic.
valueFrom	An integer. The low range value that a characteristic can take on.
valueTo	An integer. The upper range value that a characteristic can take on.
valueType	A string. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

FeatureSpecification sub-resource

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Specification for resource, service or product features.

href	An uri (Uri). Hyperlink reference.
id	A string. Identifier of the feature specification. Must be locally unique within the containing specification, thus allowing direct access to the feature spec.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureSpecCharacteristic	A list of feature specification characteristics (FeatureSpecificationCharacteristic [*]). This is a list of characteristics for a particular feature.
featureSpecRelationship	A list of feature specification relationships (FeatureSpecificationRelationship [*]). A dependency, exclusivity or aggregation relationship between/among feature specifications.
isBundle	A boolean. A flag indicating if this is a feature group (true) or not (false).
isEnabled	A boolean. A flag indicating if the feature is enabled (true) or not (false).
name	A string. Unique name given to the feature specification.
validFor	A time period. The period for which this feature specification is valid.
version	A string. Version of the feature specification.

FeatureSpecificationCharacteristic sub-resource

Configuration feature characteristic specification.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.

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isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureSpecCharRelationship	A list of feature specification characteristic relationships (FeatureSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among feature characteristics.
featureSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). Used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a FeatureCharacteristic object.
validFor	A time period. The period of time for which a characteristic is applicable.

FeatureSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among FeatureSpecificationCharacteristics.

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href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicId	A string. Unique identifier of the characteristic within the target feature specification.
featureId	A string. Unique identifier of the target feature specification within the resource specification.
name	A string. Name of the target characteristic.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target feature and feature characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target feature and feature characteristic.
validFor	A time period. The period for which the object is valid.

FeatureSpecificationRelationship sub-resource

Relationship between feature specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureId	A string. Unique identifier of the target feature specification.
name	A string. This is the name of the target feature specification.
parentSpecificationHref	An uri (Uri). Hyperlink reference to the parent specification containing the target feature.
parentSpecificationId	A string. Unique identifier of the parent specification containing the target feature.
relationshipType	A string. This is the type of the feature specification relationship.

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validFor	A time period. The period for which this feature specification relationship is valid.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Entity reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

ResourceSpecificationCharacteristic sub-resource

This class defines the characteristic features of a resource specification. Every ResourceSpecification has a variety of important attributes, methods, constraints, and
relationships, which distinguish a resource specification from other resource specifications.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.

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isUnique	A boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular JsonLdExpression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
resourceSpecCharRelationship	A list of resource specification characteristic relationships (ResourceSpecificationCharacteristicRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
resourceSpecCharacteristicValue	A list of characteristic value specifications (CharacteristicValueSpecification [*]). A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a ResourceSpecificationCharacteristic object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding ResourceSpecificationCharacteristic object can take on.

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validFor	A time period. The period of time for which a characteristic is applicable.

ResourceSpecificationCharacteristicRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among ResourceSpecificationCharacteristics. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicSpecificationId	A string. Unique identifier of the characteristic within the specification.
name	A string. Name of the target characteristic within the specification.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
resourceSpecificationHref	An uri (Uri). Hyperlink reference to the resource specification containing the target characteristic.
resourceSpecificationId	A string. Unique identifier of the resource specification containing the target characteristic.
validFor	A time period. The period for which the object is valid.

ResourceSpecificationRelationship sub-resource

A migration, substitution, dependency or exclusivity relationship between/among resource specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristic	A list of resource specification characteristics (ResourceSpecificationCharacteristic [*]). A characteristic that refines the relationship. For example, consider the relationship between a slot and a card. For a half-height card it is important to know the position at which the card is inserted, so a characteristic Position might be defined on the relationship to allow capturing of this in the inventory.

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defaultQuantity	An integer. The default number of the related resource that should be instantiated, for example a rack would typically have 4 cards, although it could support more.
href	An uri (Uri). Reference of the target ResourceSpecification.
id	A string. Unique identifier of target ResourceSpecification.
maximumQuantity	An integer. The maximum number of the related resource that should be instantiated, for example a rack supports a maximum of 16 cards.
minimumQuantity	An integer. The minimum number of the related resource that should be instantiated, for example a rack must have at least 1 card.
name	A string. The name given to the target resource specification instance.
relationshipType	A string. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A string. The association role for this resource specification.
validFor	A time period. The period for which the ResourceSpecRelationship is valid.

TargetResourceSchema sub-resource

The reference object to the schema and type of target resource which is described by resource specification.

@schemaLocation	An uri (Uri). This field provides a link to the schema describing the target resource.
@type	A string. Class type of the target resource.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

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version	A string. constraint version.

Json representation sample

Provided below is the json representation of an example of a 'SoftwareSupportPackageSpecification' resource object


```
{
  "id": "632547af-e850-45a1-a569-31531ea81eb7",
  "href": "/resourceCatalog/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7",
  "name": "Software Package Repository Specification",
  "description": "This specification describes Software Package in a repository",
  "@type": "SoftwareSupportPackageSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "SoftwareSupportPackage"
  },
  "resourceSpecRelationship": [],
  "resourceSpecCharacteristic": [
    {
      "name": "repoName",
      "description": "Name of the image repository",
      "valueType": "string"
    },
    {
      "name": "repoBase",
      "description": "Base URL of the image repository",
      "valueType": "string"
    },
    {
      "name": "imagePath",
      "description": "Path of the image in the repository",
      "valueType": "string"
    }
  ]
}
```


Notification Resource Models

7 notifications are defined for this API

Notifications related to Resource:
- ResourceCreateEvent
- ResourceAttributeValueChangeEvent

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- ResourceStateChangeEvent
- ResourceDeleteEvent

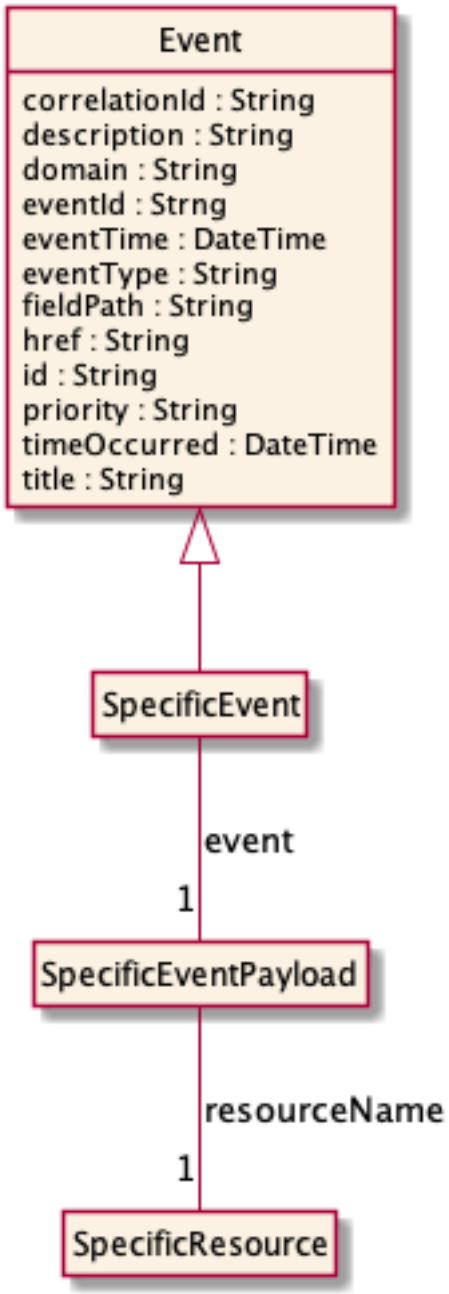
Notifications related to ResourceSpecification:
- ResourceSpecificationCreateEvent
- ResourceSpecificationChangeEvent
- ResourceSpecificationDeleteEvent

The notification structure for all notifications in this API follow the pattern depicted by the figure below.

A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType).

This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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Resource Create Event

Notification ResourceCreateEvent case for resource Resource

Json representation sample

Provided below the json representation of an example of a 'ResourceCreateEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ResourceCreateEvent",
    "event": {
```

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```
    "resource" :
      {-- SEE Resource RESOURCE SAMPLE --}
  }
}
```

Resource Attribute Value Change Event

Notification ResourceAttributeValueChangeEvent case for resource Resource

Json representation sample

Provided below is the json representation of an example of a 'ResourceAttributeValueChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ResourceAttributeValueChangeEvent",
   "event": {
     "resource" :
       {-- SEE Resource RESOURCE SAMPLE --}
  }
}
```

Resource State Change Event

Notification ResourceStateChangeEvent case for resource Resource

Json representation sample

Provided below is the json representation of an example of a 'ResourceStateChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ResourceStateChangeEvent",
   "event": {
     "resource" :
       {-- SEE Resource RESOURCE SAMPLE --}
  }
}
```

Resource Delete Event

Notification ResourceDeleteEvent case for resource Resource

Json representation sample

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Provided below is the json representation of an example of a 'ResourceDeleteEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ResourceDeleteEvent",
   "event": {
     "resource" :
       {-- SEE Resource RESOURCE SAMPLE --}
  }
}
```

Resource Specification Create Event

Notification ResourceSpecificationCreateEvent case for resource ResourceSpecification

Json representation sample

Provided below is the json representation of an example of a 'ResourceSpecificationCreateEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ResourceSpecificationCreateEvent",
   "event": {
     "resourceSpecification" :
       {-- SEE ResourceSpecification RESOURCE SAMPLE --}
  }
}
```

Resource Specification Change Event

Notification ResourceSpecificationChangeEvent case for resource ResourceSpecification

Json representation sample

Provided below is the json representation of an example of a 'ResourceSpecificationChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ResourceSpecificationChangeEvent",
   "event": {
     "resourceSpecification" :
       {-- SEE ResourceSpecification RESOURCE SAMPLE --}
  }
}
```

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Resource Specification Delete Event

Notification ResourceSpecificationDeleteEvent case for resource ResourceSpecification

Json representation sample

Provided below is the json representation of an example of a 'ResourceSpecificationDeleteEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ResourceSpecificationDeleteEvent",
    "event": {
      "resourceSpecification" :
         {-- SEE ResourceSpecification RESOURCE SAMPLE --}
   }
}
```

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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Complete Update of an Entity	PUT Resource	PUT must be used to completely update a resource identified by its resource URI
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on Resource

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List resources

`GET /resource?fields=...&{filtering}`

Description

This operation list resource entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

The following is an example of a request to list Resources:

Request

```
GET /tmf-api/softwareCompute/v4/resource
Accept: application/json
```

Response

```
200

[
  {
    "id": "6381aee0-6e5a-436f-8e20-b231d3efd712",
    "href": "/resourceInventory/v4/resource/6381aee0-6e5a-436f-8e20-b231d3efd712",
    "name": "Encabulator Software",
    "description": "This software realizes an instance of Encabulator",
    "@type": "InstalledSoftware",
    "startOperatingDate": "2022-11-28T08:00:00Z",
    "endOperatingDate": "2023-11-29T00:00:00Z",
    "lastUpdate": "2022-02-04T15:10:05Z",
    "category": "Software",
    "resourceSpecification": {
      "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
      "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
      "name": "Encabulator Software Specification",
      "@referredType": "SoftwareSpecification"
    },
    "isUTCTime": true,
    "lastStartTime": "2022-02-04T15:10:01Z",
    "targetPlatform": "server",
    "numProcessesActiveCurrent": 1,
    "numUsersCurrent": 0,
    "pagingFileSizeCurrent": {
      "amount": 49152,
      "units": "4 KiB"
    },
```

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```
    "processMemorySizeCurrent": {
      "amount": 1399714284,
      "units": "bytes"
    },
    "swapSpaceUsedCurrent": {
      "amount": 0,
      "units": "bytes"
    },
    "serialNumber": "831038429801",
    "resourceRelationship": [
      {
        "relationshipType": "uses",
        "resource": {
          "id": "a1976344-3933-4942-874a-ded80c1e5d83",
          "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
          "name": "Encabulator Platform Requirement",
          "@referredType": "HostingPlatformRequirement"
        }
      },
      {
        "relationshipType": "provides",
        "resource": {
          "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
          "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
          "name": "Encabulator Function",
          "@referredType": "ResourceFunction"
        }
      }
    ],
    "resourceCharacteristic": [
      {
        "name": "OperatingSystem",
        "value": "Debian 11 (bullseye)"
      }
    ]
  },
  {
    "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
    "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
    "name": "Encabulator Function",
    "description": "The Encabulator function.",
    "@type": "ResourceFunction",
    "startOperatingDate": "2022-11-28T08:00:00Z",
    "endOperatingDate": "2023-11-29T00:00:00Z",
    "category": "Software",
    "resourceSpecification": {
      "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "name": "Encabulator Function Specification",
      "@referredType": "ResourceFunctionSpecification"
    },
    "resourceRelationship": [],
    "connectionPoint": [
      {
        "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
```


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```
        "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
        "name": "TMF635",
        "@referredType": "API"
      }
    ],
    "connectivity": [],
    "activationFeature": [],
    "resourceCharacteristic": []
  },
  {
    "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
    "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
    "name": "TMF635",
    "description": "Open API for Usage Management",
    "@type": "API",
    "startOperatingDate": "2021-07-01T08:00:00Z",
    "category": "Software",
    "resourceSpecification": {
      "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
      "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
      "name": "TMF635 Specification",
      "@referredType": "APISpecification"
    },
    "resourceRelationship": [],
    "resourceCharacteristic": []
  },
  {
    "id": "a1976344-3933-4942-874a-ded80c1e5d83",
    "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
    "name": "Encabulator Platform Requirement",
    "description": "The requirements on platforms hosting the Encabulator software",
    "@type": "HostingPlatformRequirement",
    "startOperatingDate": "2022-11-28T08:00:00Z",
    "endOperatingDate": "2023-11-29T00:00:00Z",
    "category": "Compute",
    "resourceSpecification": {
      "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "name": "Encabulator Platform Requirement Specification",
      "@referredType": "HostingPlatformRequirementSpecification"
    },
    "resourceRelationship": [],
    "activationFeature": [
      {
        "name": "scale",
        "featureCharacteristic": [
          {
            "name": "cpuSize",
            "value": "micro"
          },
          {
            "name": "cpuNum",
            "value": 2
          },
          {
```


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```
                "name": "memory",
                "value": 1024
              }
            ]
          }
        ],
        "resourceCharacteristic": [
          {
            "name": "virtualization",
            "value": "container-runtime"
          }
        ]
      },
      {
        "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
        "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
        "name": "Encabulator Software Package",
        "description": "Package providing the files containing the Encabulator software",
        "@type": "SoftwareSupportPackage",
        "startOperatingDate": "2022-11-28T08:00:00Z",
        "endOperatingDate": "2023-11-29T00:00:00Z",
        "category": "Software",
        "resourceSpecification": {
          "id": "632547af-e850-45a1-a569-31531ea81eb7",
          "href": "/resourceCatalog/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7",
          "name": "Software Package Repository Specification"
        },
        "resourceRelationship": [],
        "resourceCharacteristic": [
          {
            "name": "repoName",
            "value": "Local"
          },
          {
            "name": "repoBase",
            "value": "https://repo.example.net"
          },
          {
            "name": "imagePath",
            "value": "/cnf/9c53e70b-5e78-41e8-826b-8c70c2b65a8e"
          }
        ]
      }
    ]
```


Retrieve resource

`GET /resource/{id}?fields=...&{filtering}`

Description

This operation retrieves a resource entity.
Attribute selection is enabled for all first level attributes.

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Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

The following is an example of a request to get an InstalledSoftware:

Request

```
GET /tmf-api/softwareCompute/v4/resource/6381aee0-6e5a-436f-8e20-b231d3efd712
Accept: application/json
```

Response

```
200

{
  "id": "6381aee0-6e5a-436f-8e20-b231d3efd712",
  "href": "/resourceInventory/v4/resource/6381aee0-6e5a-436f-8e20-b231d3efd712",
  "name": "Encabulator Software",
  "description": "This software realizes an instance of Encabulator",
  "@type": "InstalledSoftware",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "lastUpdate": "2022-02-04T15:10:05Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "name": "Encabulator Software Specification",
    "@referredType": "SoftwareSpecification"
  },
  "isUTCTime": true,
  "lastStartTime": "2022-02-04T15:10:01Z",
  "targetPlatform": "server",
  "numProcessesActiveCurrent": 1,
  "numUsersCurrent": 0,
  "pagingFileSizeCurrent": {
    "amount": 49152,
    "units": "4 KiB"
  },
  "processMemorySizeCurrent": {
    "amount": 1399714284,
    "units": "bytes"
  },
  "swapSpaceUsedCurrent": {
    "amount": 0,
    "units": "bytes"
  },
```

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```
  "serialNumber": "831038429801",
  "resourceRelationship": [
    {
      "relationshipType": "uses",
      "resource": {
        "id": "a1976344-3933-4942-874a-ded80c1e5d83",
        "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
        "name": "Encabulator Platform Requirement",
        "@referredType": "HostingPlatformRequirement"
      }
    },
    {
      "relationshipType": "provides",
      "resource": {
        "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "name": "Encabulator Function",
        "@referredType": "ResourceFunction"
      }
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "OperatingSystem",
      "value": "Debian 11 (bullseye)"
    }
  ]
}
```


The following is an example of a request to get a ResourceFunction:

Request

```
GET /tmf-api/softwareCompute/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77
Accept: application/json
```

Response

200


```
{
  "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
  "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
  "name": "Encabulator Function",
  "description": "The Encabulator function.",
  "@type": "ResourceFunction",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
```


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```
      "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "name": "Encabulator Function Specification",
      "@referredType": "ResourceFunctionSpecification"
   },
   "resourceRelationship": [],
   "connectionPoint": [
      {
         "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
         "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
         "name": "TMF635",
         "@referredType": "API"
      }
   ],
   "connectivity": [],
   "activationFeature": [],
   "resourceCharacteristic": []
}
```


The following is an example of a request to get an API:

Request

```
GET /tmf-api/softwareCompute/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856
Accept: application/json
```

Response

```
200

{
   "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
   "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
   "name": "TMF635",
   "description": "Open API for Usage Management",
   "@type": "API",
   "startOperatingDate": "2021-07-01T08:00:00Z",
   "lastUpdate": "2022-11-27T20:51:14.457Z",
   "category": "Software",
   "resourceSpecification": {
      "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
      "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
      "name": "TMF635 Specification",
      "@referredType": "APISpecification"
   },
   "resourceRelationship": [],
   "resourceCharacteristic": []
}
```

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The following is an example of a request to get a HostingPlatformRequirement:

Request

```
GET /tmf-api/softwareCompute/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83
Accept: application/json
```

Response

```
200

{
  "id": "a1976344-3933-4942-874a-ded80c1e5d83",
  "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
  "name": "Encabulator Platform Requirement",
  "description": "The requirements on platforms hosting the Encabulator software",
  "@type": "HostingPlatformRequirement",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Compute",
  "resourceSpecification": {
    "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
    "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
    "name": "Encabulator Platform Requirement Specification",
    "@referredType": "HostingPlatformRequirementSpecification"
  },
  "resourceRelationship": [],
  "activationFeature": [
    {
      "name": "scale",
      "featureCharacteristic": [
        {
          "name": "cpuSize",
          "value": "micro"
        },
        {
          "name": "cpuNum",
          "value": 2
        },
        {
          "name": "memory",
          "value": 1024
        }
      ]
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "virtualization",
      "value": "container-runtime"
```

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```
    }
  ]
}
```

The following is an example of a request to get a SoftwareSupportPackage:

Request

```
GET /tmf-api/softwareCompute/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7
Accept: application/json
```

Response

```
200

{
  "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
  "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
  "name": "Encabulator Software Package",
  "description": "Package providing the files containing the Encabulator software",
  "@type": "SoftwareSupportPackage",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "632547af-e850-45a1-a569-31531ea81eb7",
    "href": "/resourceCatalog/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7",
    "name": "Software Package Repository Specification"
  },
  "resourceRelationship": [],
  "resourceCharacteristic": [
    {
      "name": "repoName",
      "value": "Local"
    },
    {
      "name": "repoBase",
      "value": "https://repo.example.net"
    },
    {
      "name": "imagePath",
      "value": "/cnf/9c53e70b-5e78-41e8-826b-8c70c2b65a8e"
    }
  ]
}
```

Create resource

POST /resource

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Description

This operation creates a resource entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a Resource, including any possible rule conditions and applicable default values. Notice that it is up to the implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
name	
@type	
href	
id	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
activationFeature	
administrativeState	
attachment	
category	
description	
endOperatingDate	
note	
operationalState	
place	
relatedParty	
resourceCharacteristic	
resourceRelationship	
resourceSpecification	
resourceStatus	
resourceVersion	
startOperatingDate	
usageState	

Usage Samples

The following is an example of a request to create an InstalledSoftware:

Request

```
POST /tmf-api/softwareCompute/v4/resource
Content-Type: application/json

{
   "name": "Encabulator Software",
   "description": "This software realizes an instance of Encabulator",
```

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```
  "@type": "InstalledSoftware",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatinghDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "name": "Encabulator Software Specification",
    "@referredType": "SoftwareSpecification"
  },
  "targetPlatform": "server",
  "serialNumber": "831038429801",
  "resourceRelationship": [
    {
      "relationshipType": "uses",
      "resource": {
        "id": "a1976344-3933-4942-874a-ded80c1e5d83",
        "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
        "name": "Encabulator Platform Requirement",
        "@referredType": "HostingPlatformRequirement"
      }
    },
    {
      "relationshipType": "provides",
      "resource": {
        "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "name": "Encabulator Function",
        "@referredType": "ResourceFunction"
      }
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "OperatingSystem",
      "value": "Debian 11 (bullseye)"
    }
  ]
}
```


Response

201


```
{
  "id": "6381aee0-6e5a-436f-8e20-b231d3efd712",
  "href": "/resourceInventory/v4/resource/6381aee0-6e5a-436f-8e20-b231d3efd712",
  "name": "Encabulator Software",
  "description": "This software realizes an instance of Encabulator",
  "@type": "InstalledSoftware",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "lastUpdate": "2022-02-04T15:10:05Z",
```


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```
  "category": "Software",
  "resourceSpecification": {
    "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "name": "Encabulator Software Specification",
    "@referredType": "SoftwareSpecification"
  },
  "isUTCTime": true,
  "lastStartTime": "2022-02-04T15:10:01Z",
  "targetPlatform": "server",
  "numProcessesActiveCurrent": 1,
  "numUsersCurrent": 0,
  "pagingFileSizeCurrent": {
    "amount": 49152,
    "units": "4 KiB"
  },
  "processMemorySizeCurrent": {
    "amount": 1399714284,
    "units": "bytes"
  },
  "swapSpaceUsedCurrent": {
    "amount": 0,
    "units": "bytes"
  },
  "serialNumber": "831038429801",
  "resourceRelationship": [
    {
      "relationshipType": "uses",
      "resource": {
        "id": "a1976344-3933-4942-874a-ded80c1e5d83",
        "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
        "name": "Encabulator Platform Requirement",
        "@referredType": "HostingPlatformRequirement"
      }
    },
    {
      "relationshipType": "provides",
      "resource": {
        "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "name": "Encabulator Function",
        "@referredType": "ResourceFunction"
      }
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "OperatingSystem",
      "value": "Debian 11 (bullseye)"
    }
  ]
}
```


The following is an example of a request to create a ResourceFunction:

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Request

```
POST /tmf-api/softwareCompute/v4/resource
Content-Type: application/json

{
  "name": "Encabulator Function",
  "description": "The Encabulator function.",
  "@type": "ResourceFunction",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "name": "Encabulator Function Specification",
    "@referredType": "ResourceFunctionSpecification"
  },
  "connectionPoint": [
    {
      "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
      "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
      "name": "TMF635",
      "@referredType": "API"
    }
  ]
}
```

Response

```
201

{
  "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
  "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
  "name": "Encabulator Function",
  "description": "The Encabulator function.",
  "@type": "ResourceFunction",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "name": "Encabulator Function Specification",
    "@referredType": "ResourceFunctionSpecification"
  },
  "resourceRelationship": [],
  "connectionPoint": [
    {
      "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
```

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```
        "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
        "name": "TMF635",
        "@referredType": "API"
      }
   ],
   "connectivity": [],
   "activationFeature": [],
   "resourceCharacteristic": []
}
```


The following is an example of a request to create an API:

Request

```
POST /tmf-api/softwareCompute/v4/resource
Content-Type: application/json

{
   "name": "TMF635",
   "description": "Open API for Usage Management",
   "@type": "API",
   "startOperatingDate": "2021-07-01T08:00:00Z",
   "category": "Software",
   "resourceSpecification": {
      "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
      "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
      "name": "TMF635 Specification",
      "@referredType": "APISpecification"
   }
}
```

Response


```
201

{
   "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
   "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
   "name": "TMF635",
   "description": "Open API for Usage Management",
   "@type": "API",
   "startOperatingDate": "2021-07-01T08:00:00Z",
   "lastUpdate": "2022-11-27T20:51:14.457Z",
   "category": "Software",
   "resourceSpecification": {
      "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
      "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
      "name": "TMF635 Specification",
      "@referredType": "APISpecification"
   },
```


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```
  "resourceRelationship": [],
  "resourceCharacteristic": []
}
```

The following is an example of a request to create a HostingPlatformRequirement:

Request

```
POST /tmf-api/softwareCompute/v4/resource
Content-Type: application/json

{
  "name": "Encabulator Platform Requirement",
  "description": "The requirements on platforms hosting the Encabulator software",
  "@type": "HostingPlatformRequirement",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Compute",
  "resourceSpecification": {
    "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
    "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
    "name": "Encabulator Platform Requirement Specification",
    "@referredType": "HostingPlatformRequirementSpecification"
  },
  "activationFeature": [
    {
      "name": "scale",
      "featureCharacteristic": [
        {
          "name": "cpuSize",
          "value": "micro"
        },
        {
          "name": "cpuNum",
          "value": 2
        },
        {
          "name": "memory",
          "value": 1024
        }
      ]
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "virtualization",
      "value": "container-runtime"
    }
  ]
}
```

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Response

```
201

{
  "id": "a1976344-3933-4942-874a-ded80c1e5d83",
  "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
  "name": "Encabulator Platform Requirement",
  "description": "The requirements on platforms hosting the Encabulator software",
  "@type": "HostingPlatformRequirement",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Compute",
  "resourceSpecification": {
    "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
    "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
    "name": "Encabulator Platform Requirement Specification",
    "@referredType": "HostingPlatformRequirementSpecification"
  },
  "resourceRelationship": [],
  "activationFeature": [
    {
      "name": "scale",
      "featureCharacteristic": [
        {
          "name": "cpuSize",
          "value": "micro"
        },
        {
          "name": "cpuNum",
          "value": 2
        },
        {
          "name": "memory",
          "value": 1024
        }
      ]
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "virtualization",
      "value": "container-runtime"
    }
  ]
}
```

The following is an example of a request to create an SoftwareSupportPackage:

Request

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```
POST /tmf-api/softwareCompute/v4/resource
Content-Type: application/json

{
  "name": "Encabulator Software Package",
  "description": "Package providing the files containing the Encabulator software",
  "@type": "SoftwareSupportPackage",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "632547af-e850-45a1-a569-31531ea81eb7",
    "href": "/resourceCatalog/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7",
    "name": "Software Package Repository Specification"
  },
  "resourceCharacteristic": [
    {
      "name": "repoName",
      "value": "Local"
    },
    {
      "name": "repoBase",
      "value": "https://repo.example.net"
    },
    {
      "name": "imagePath",
      "value": "/cnf/9c53e70b-5e78-41e8-826b-8c70c2b65a8e"
    }
  ]
}
```

Response

```
201

{
  "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
  "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
  "name": "Encabulator Software Package",
  "description": "Package providing the files containing the Encabulator software",
  "@type": "SoftwareSupportPackage",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "632547af-e850-45a1-a569-31531ea81eb7",
    "href": "/resourceCatalog/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7",
    "name": "Software Package Repository Specification"
  },
  "resourceRelationship": [],
  "resourceCharacteristic": [
    {
      "name": "repoName",
```

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```
        "value": "Local"
      },
      {
        "name": "repoBase",
        "value": "https://repo.example.net"
      },
      {
        "name": "imagePath",
        "value": "/cnf/9c53e70b-5e78-41e8-826b-8c70c2b65a8e"
      }
    ]
}
```


Patch resource

`PATCH /resource/{id}`

Description

This operation allows partial updates of a resource entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non-patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
@baseType	
@schemaLocation	
activationFeature	
administrativeState	
attachment	
category	
description	
endOperatingDate	
name	
note	
operationalState	
place	
relatedParty	
resourceCharacteristic	
resourceRelationship	
resourceSpecification	
resourceStatus	

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resourceVersion	
startOperatingDate	
usageState	

Non Patchable Attributes	Rule
id	
href	
@type	

Usage Samples

Example of updating an InstalledSoftware using json-patch:

Request

```
PATCH /tmf-api/softwareCompute/v4/resource/6381aee0-6e5a-436f-8e20-b231d3efd712
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/numProcessesActiveCurrent",
    "value": 2
  },
  {
    "op": "replace",
    "path": "/numProcessesActiveCurrent",
    "value": 1
  },
  {
    "op": "replace",
    "path": "/pagingFileSizeCurrent",
    "value": 50768
  },
  {
    "op": "replace",
    "path": "/processMemorySizeCurrent",
    "value": 1403281196
  }
]
```

Response

```
200

{
  "id": "6381aee0-6e5a-436f-8e20-b231d3efd712",
  "href": "/resourceInventory/v4/resource/6381aee0-6e5a-436f-8e20-b231d3efd712",
  "name": "Encabulator Software",
```

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```
  "description": "This software realizes an instance of Encabulator",
  "@type": "InstalledSoftware",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "lastUpdate": "2022-11-28T15:10:05Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
    "name": "Encabulator Software Specification",
    "@referredType": "SoftwareSpecification"
  },
  "isUTCTime": true,
  "lastStartTime": "2022-02-04T15:10:01Z",
  "targetPlatform": "server",
  "numProcessesActiveCurrent": 2,
  "numUsersCurrent": 1,
  "pagingFileSizeCurrent": {
    "amount": 50768,
    "units": "4 KiB"
  },
  "processMemorySizeCurrent": {
    "amount": 1403281196,
    "units": "bytes"
  },
  "swapSpaceUsedCurrent": {
    "amount": 0,
    "units": "bytes"
  },
  "serialNumber": "831038429801",
  "resourceRelationship": [
    {
      "relationshipType": "uses",
      "resource": {
        "id": "a1976344-3933-4942-874a-ded80c1e5d83",
        "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
        "name": "Encabulator Platform Requirement",
        "@referredType": "HostingPlatformRequirement"
      }
    },
    {
      "relationshipType": "provides",
      "resource": {
        "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
        "name": "Encabulator Function",
        "@referredType": "ResourceFunction"
      }
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "OperatingSystem",
      "value": "Debian 11 (bullseye)"
    }
```


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```
  ]
}
```

Example of updating a ResourceFunction using json-patch:

Request

```
PATCH /tmf-api/softwareCompute/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77
Content-Type: application/json-patch+json

[
  {
    "op": "add",
    "path": "/resourceCharacteristic/-",
    "value": {
      "name": "numMarzelvanes",
      "value": 6
    }
  },
  {
    "op": "add",
    "path": "/resourceCharacteristic/-",
    "value": {
      "name": "numStatorSlots",
      "value": 47
    }
  }
]
```

Response

```
200

{
  "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
  "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
  "name": "Encabulator Function",
  "description": "The Encabulator function.",
  "@type": "ResourceFunction",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "name": "Encabulator Function Specification",
    "@referredType": "ResourceFunctionSpecification"
  },
  "resourceRelationship": [],
  "connectionPoint": [
```

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```
    {
      "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
      "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
      "name": "TMF635",
      "@referredType": "API"
    }
  ],
  "connectivity": [],
  "activationFeature": [],
  "resourceCharacteristic": [
    {
      "name": "numMarzelvanes",
      "value": 6
    },
    {
      "name": "numStatorSlots",
      "value": 47
    }
  ]
}
```


Example of updating an API using json-patch:

Request


```
PATCH /tmf-api/softwareCompute/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856
Content-Type: application/json-patch+json

[
  {
    "op": "add",
    "path": "/resourceCharacteristic/-",
    "value": {
      "name": "numMarzelvanes",
      "value": 6
    }
  },
  {
    "op": "add",
    "path": "/resourceCharacteristic/-",
    "value": {
      "name": "numStatorSlots",
      "value": 47
    }
  }
]
```


Response

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```
200

{
  "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
  "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
  "name": "Encabulator Function",
  "description": "The Encabulator function.",
  "@type": "ResourceFunction",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "name": "Encabulator Function Specification",
    "@referredType": "ResourceFunctionSpecification"
  },
  "resourceRelationship": [],
  "connectionPoint": [
    {
      "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
      "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
      "name": "TMF635",
      "@referredType": "API"
    }
  ],
  "connectivity": [],
  "activationFeature": [],
  "resourceCharacteristic": [
    {
      "name": "numMarzelvanes",
      "value": 6
    },
    {
      "name": "numStatorSlots",
      "value": 47
    }
  ]
}
```

Example of updating a HostingPlatformRequirement using json-patch:

Request

```
PATCH /tmf-api/softwareCompute/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/activationFeature/0/featureCharacteristic/0/value",
    "value": "small"
```

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```
  },
  {
    "op": "replace",
    "path": "/activationFeature/0/featureCharacteristic/2/value",
    "value": 1536
  }
]
```

Response

200

```
{
  "id": "a1976344-3933-4942-874a-ded80c1e5d83",
  "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
  "name": "Encabulator Platform Requirement",
  "description": "The requirements on platforms hosting the Encabulator software",
  "@type": "HostingPlatformRequirement",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Compute",
  "resourceSpecification": {
    "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
    "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
    "name": "Encabulator Platform Requirement Specification",
    "@referredType": "HostingPlatformRequirementSpecification"
  },
  "resourceRelationship": [],
  "activationFeature": [
    {
      "name": "scale",
      "featureCharacteristic": [
        {
          "name": "cpuSize",
          "value": "small"
        },
        {
          "name": "cpuNum",
          "value": 2
        },
        {
          "name": "memory",
          "value": 1536
        }
      ]
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "virtualization",
      "value": "container-runtime"
    }
  ]
```

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```
}
```

Example of updating a SoftwareSupportPackage using json-patch:

Request

```
PATCH /tmf-api/softwareCompute/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/resourceCharacteristic/2/value",
    "value": "/cnf/d49f5802-359e-424e-a97b-3cd22f0bee2d"
  }
]
```

Response

```
200

{
  "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
  "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
  "name": "Encabulator Software Package",
  "description": "Package providing the files containing the Encabulator software",
  "@type": "SoftwareSupportPackage",
  "startOperatingDate": "2022-11-28T08:00:00Z",
  "endOperatingDate": "2023-11-29T00:00:00Z",
  "category": "Software",
  "resourceSpecification": {
    "id": "632547af-e850-45a1-a569-31531ea81eb7",
    "href": "/resourceCatalog/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7",
    "name": "Software Package Repository Specification"
  },
  "resourceRelationship": [],
  "resourceCharacteristic": [
    {
      "name": "repoName",
      "value": "Local"
    },
    {
      "name": "repoBase",
      "value": "https://repo.example.net"
    },
    {
      "name": "imagePath",
      "value": "/cnf/d49f5802-359e-424e-a97b-3cd22f0bee2d"
    }
  ]
```

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```
}
```

Delete resource

`DELETE /resource/{id}`

Description

This operation deletes a resource entity.

Usage Samples

Example of a request for deleting an InstalledSoftware:

Request
DELETE /tmf-api/softwareCompute/v4/resource/6381aee0-6e5a-436f-8e20-b231d3efd712
Response
204

Example of a request for deleting a ResourceFunction:

Request
DELETE /tmf-api/softwareCompute/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77
Response
204

Example of a request for deleting an API:

Request
DELETE /tmf-api/softwareCompute/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856

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Response
204

Example of a request for deleting a HostingPlatformRequirement:

Request
DELETE /tmf-api/softwareCompute/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83
Response
204

Example of a request for deleting a SoftwareSupportPackage:

Request
DELETE /tmf-api/softwareCompute/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7
Response
204

Operations on Resource Specification

List resource specifications

```
GET /resourceSpecification?fields=...&{filtering}
```

Description

This operation list resource specification entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

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The following is an example of a request to list ResourceSspecifications:

Request

```
GET /tmf-api/softwareCompute/v4/resourceSpecification
Accept: application/json
```

Response

```
200

[
  {
    "id": "6381aee0-6e5a-436f-8e20-b231d3efd712",
    "href": "/resourceInventory/v4/resource/6381aee0-6e5a-436f-8e20-b231d3efd712",
    "name": "Encabulator Software",
    "description": "This software realizes an instance of Encabulator",
    "@type": "InstalledSoftware",
    "startOperatingDate": "2022-11-28T08:00:00Z",
    "endOperatingDate": "2023-11-29T00:00:00Z",
    "lastUpdate": "2022-02-04T15:10:05Z",
    "category": "Software",
    "resourceSpecification": {
      "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
      "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
      "name": "Encabulator Software Specification",
      "@referredType": "SoftwareSpecification"
    },
    "isUTCTime": true,
    "lastStartTime": "2022-02-04T15:10:01Z",
    "targetPlatform": "server",
    "numProcessesActiveCurrent": 1,
    "numUsersCurrent": 0,
    "pagingFileSizeCurrent": {
      "amount": 49152,
      "units": "4 KiB"
    },
    "processMemorySizeCurrent": {
      "amount": 1399714284,
      "units": "bytes"
    },
    "swapSpaceUsedCurrent": {
      "amount": 0,
      "units": "bytes"
    },
    "serialNumber": "831038429801",
    "resourceRelationship": [
      {
        "relationshipType": "uses",
        "resource": {
          "id": "a1976344-3933-4942-874a-ded80c1e5d83",
```

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```
            "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
            "name": "Encabulator Platform Requirement",
            "@referredType": "HostingPlatformRequirement"
          }
        },
        {
          "relationshipType": "provides",
          "resource": {
            "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
            "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
            "name": "Encabulator Function",
            "@referredType": "ResourceFunction"
          }
        }
      ],
      "resourceCharacteristic": [
        {
          "name": "OperatingSystem",
          "value": "Debian 11 (bullseye)"
        }
      ]
    },
    {
      "id": "03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
      "href": "/resourceInventory/v4/resource/03c5a7d1-6cd5-4467-b0f5-fe7c06fb5e77",
      "name": "Encabulator Function",
      "description": "The Encabulator function.",
      "@type": "ResourceFunction",
      "startOperatingDate": "2022-11-28T08:00:00Z",
      "endOperatingDate": "2023-11-29T00:00:00Z",
      "category": "Software",
      "resourceSpecification": {
        "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
        "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
        "name": "Encabulator Function Specification",
        "@referredType": "ResourceFunctionSpecification"
      },
      "resourceRelationship": [],
      "connectionPoint": [
        {
          "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
          "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
          "name": "TMF635",
          "@referredType": "API"
        }
      ],
      "connectivity": [],
      "activationFeature": [],
      "resourceCharacteristic": []
    },
    {
      "id": "87839f7e-e490-4cfb-b371-6d1edeb85856",
      "href": "/resourceInventory/v4/resource/87839f7e-e490-4cfb-b371-6d1edeb85856",
      "name": "TMF635",
      "description": "Open API for Usage Management",
```


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```
    "@type": "API",
    "startOperatingDate": "2021-07-01T08:00:00Z",
    "category": "Software",
    "resourceSpecification": {
      "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
      "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
      "name": "TMF635 Specification",
      "@referredType": "APISpecification"
    },
    "resourceRelationship": [],
    "resourceCharacteristic": []
  },
  {
    "id": "a1976344-3933-4942-874a-ded80c1e5d83",
    "href": "/resourceInventory/v4/resource/a1976344-3933-4942-874a-ded80c1e5d83",
    "name": "Encabulator Platform Requirement",
    "description": "The requirements on platforms hosting the Encabulator software",
    "@type": "HostingPlatformRequirement",
    "startOperatingDate": "2022-11-28T08:00:00Z",
    "endOperatingDate": "2023-11-29T00:00:00Z",
    "category": "Compute",
    "resourceSpecification": {
      "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "name": "Encabulator Platform Requirement Specification",
      "@referredType": "HostingPlatformRequirementSpecification"
    },
    "resourceRelationship": [],
    "activationFeature": [
      {
        "name": "scale",
        "featureCharacteristic": [
          {
            "name": "cpuSize",
            "value": "micro"
          },
          {
            "name": "cpuNum",
            "value": 2
          },
          {
            "name": "memory",
            "value": 1024
          }
        ]
      }
    ],
    "resourceCharacteristic": [
      {
        "name": "virtualization",
        "value": "container-runtime"
      }
    ]
  },
  {
```


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```
    "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
    "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
    "name": "Encabulator Software Package",
    "description": "Package providing the files containing the Encabulator software",
    "@type": "SoftwareSupportPackage",
    "startOperatingDate": "2022-11-28T08:00:00Z",
    "endOperatingDate": "2023-11-29T00:00:00Z",
    "category": "Software",
    "resourceSpecification": {
      "id": "632547af-e850-45a1-a569-31531ea81eb7",
      "href": "/resourceCatalog/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7",
      "name": "Software Package Repository Specification"
    },
    "resourceRelationship": [],
    "resourceCharacteristic": [
      {
        "name": "repoName",
        "value": "Local"
      },
      {
        "name": "repoBase",
        "value": "https://repo.example.net"
      },
      {
        "name": "imagePath",
        "value": "/cnf/9c53e70b-5e78-41e8-826b-8c70c2b65a8e"
      }
    ]
  }
]
```


Retrieve resource specification

`GET /resourceSpecification/{id}?fields=...&{filtering}`

Description

This operation retrieves a resource specification entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

The following is an example of a request to get a SoftwareSpecification:

Request

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```
GET /tmf-api/softwareCompute/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4
Accept: application/json
```

Response

```
200

{
  "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
  "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
  "name": "Encabulator Software Specification",
  "description": "This specification describes the Encabulator software",
  "@type": "SoftwareSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lastUpdate": "2022-02-03T07:59:59Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "InstalledSoftware"
  },
  "relatedParty": [
    {
      "id": "ab5660aa-064d-440c-b7d1-b943e42b904e",
      "href": "/partyManagement/v4/organization/ab5660aa-064d-440c-b7d1-b943e42b904e",
      "role": "supplier",
      "name": "General Electric"
    }
  ],
  "releaseStatus": "firstCustomerShipment",
  "buildNumber": "encabulator-1.0.0",
  "majorVersion": "1",
  "minorVersion": "0",
  "maintenanceVersion": "0",
  "otherDesignator": "alpha",
  "isDistributable": true,
  "installSize": {
    "amount": 9143256,
    "units": "bytes"
  },
  "isExperimental": false,
  "numUsersMax": 8,
  "numberProcessActiveTotal": 0,
  "resourceSpecRelationship": [
    {
      "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "name": "Encabulator Platform Requirement Specification",
      "relationshipType": "requires"
    },
```

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```
    {
      "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "name": "Encabulator Function Specification",
      "relationshipType": "provides"
    }
  ],
  "softwareSupportPackage": {
    "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
    "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
    "name": "Encabulator Software Package",
    "@referredType": "SoftwareSupportPackage"
  },
  "resourceSpecCharacteristic": [
    {
      "name": "OperatingSystem",
      "description": "Operating System",
      "valueType": "String",
      "configurable": true,
      "@type": "ResourceSpecificationCharacteristic",
      "minCardinality": 1,
      "maxCardinality": 3,
      "isUnique": true,
      "extensible": false,
      "resourceSpecCharacteristicValue": [
        {
          "isDefault": true,
          "value": "Debian 11 (bullseye)"
        },
        {
          "isDefault": false,
          "value": "Ubuntu 20.04 LTS (focal)"
        },
        {
          "isDefault": false,
          "value": "RHEL 9"
        }
      ]
    }
  ]
}
```


The following is an example of a request to get a ResourceFunctionSpecification:

Request

```
GET /tmf-api/softwareCompute/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241
Accept: application/json
```

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Response

200


```
{
  "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
  "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
  "name": "Encabulator Function Specification",
  "description": "This specification describes the Encabulator Function",
  "@type": "ResourceFunctionSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lastUpdate": "2022-02-03T07:59:59Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "ResourceFunction"
  },
  "resourceSpecRelationship": [],
  "featureSpecification": [],
  "resourceSpecCharacteristic": [
    {
      "name": "numMarzelvanes",
      "description": "Number of antigravic marzelvanes.",
      "valueType": "integer",
      "resourceSpecCharacteristicValue": [
        {
          "valueFrom": 6,
          "valueTo": 32
        }
      ]
    },
    {
      "name": "numStatorSlots",
      "description": "Number of panandermic semi-boloid stator slots.",
      "valueType": "integer",
      "resourceSpecCharacteristicValue": [
        {
          "valueFrom": 0,
          "valueTo": 50
        }
      ]
    }
  ],
  "connectionPointSpecification": [
    {
      "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
      "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
      "name": "TMF635 Specification",
      "@referredType": "APISpecification"
    }
```


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```
  ],
  "connectivitySpecification": []
}
```


The following is an example of a request to get an APISpecification:

Request

```
GET /tmf-api/softwareCompute/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8
Accept: application/json
```

Response


```
200

{
  "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
  "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
  "name": "TMF635 Specification",
  "description": "Open API for Usage Management",
  "@type": "APISpecification",
  "validFor": {
    "startDateTime": "2021-07-01T08:00:00Z"
  },
  "lifecycleStatus": "Active",
  "lastUpdate": "2022-11-27T07:59:59Z",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "API"
  },
  "releaseStatus": "generalDeployment",
  "buildNumber": "20.5.0",
  "majorVersion": "4",
  "minorVersion": "0",
  "maintenanceVersion": "0",
  "isDistributable": true,
  "isExperimental": false,
  "apiProtocolType": "REST",
  "externalUrl": "https://example.net/tmf-api/usageManagement/v4",
  "internalUrl": "/tmf-api/usageManagement/v4",
  "externalSchema": "https://tmf-open-api-table-documents.s3.eu-west-
1.amazonaws.com/OpenAPITable/4.0.0/swagger/TMF635-UsageManagement-v4.0.0.swagger.json",
  "internalSchema": "/tmf-api/usageManagement/v4/schema/TMF635-UsageManagement-v4.0.0.swagger.json",
  "authenticationType": "Basic",
  "allowedOperations": [
    "GET",
    "HEAD",
    "POST",
    "PATCH",
```


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```
      "DELETE"
   ],
   "allowedAPIEntities": [
      "usageSpecification",
      "usage"
   ],
   "responseTypeFormat": [
      "application/json",
      "application/problem+json"
   ],
   "resourceSpecRelationship": [],
   "relatedParty": [
      {
         "id": "ab5660aa-064d-440c-b7d1-b943e42b904e",
         "href": "/partyManagement/v4/organization/ab5660aa-064d-440c-b7d1-b943e42b904e",
         "name": "TM Forum",
         "role": "supplier"
      }
   ],
   "resourceSpecCharacteristic": []
}
```


The following is an example of a request to get a HostingPlatformRequirementSpecification:

Request

```
GET /tmf-api/softwareCompute/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389
Accept: application/json
```

Response


```
200

{
   "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
   "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
   "name": "Encabulator Platform Requirement Specification",
   "description": "This specification describes requirements on platforms hosting the Encabulator software",
   "@type": "HostingPlatformRequirementSpecification",
   "validFor": {
      "startDateTime": "2022-11-28T08:00:00Z",
      "endDateTime": "2023-11-29T00:00:00Z"
   },
   "lifecycleStatus": "Active",
   "isBundle": false,
   "category": "Compute",
   "targetResourceSchema": {
      "@type": "HostingPlatformRequirement"
   },
   "resourceSpecRelationship": [],
```


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```
"featureSpecification": [
  {
    "name": "scale",
    "featureSpecCharacteristic": [
      {
        "name": "cpuSize",
        "description": "Vertical CPU scaling dimension",
        "valueType": "string",
        "featureSpecCharacteristicValue": [
          {
            "isDefault": true,
            "value": "micro"
          },
          {
            "isDefault": false,
            "value": "small"
          },
          {
            "isDefault": false,
            "value": "medium"
          },
          {
            "isDefault": false,
            "value": "large"
          }
        ]
      },
      {
        "name": "cpuNum",
        "description": "Minimum number of CPUs",
        "valueType": "integer",
        "featureSpecCharacteristicValue": [
          {
            "valueFrom": 1,
            "valueTo": 24
          }
        ]
      },
      {
        "name": "memory",
        "description": "Minimum amount of memory required",
        "valueType": "integer",
        "featureSpecCharacteristicValue": [
          {
            "unitOfMeasure": "MiB",
            "valueFrom": 640,
            "valueTo": 16384
          }
        ]
      }
    ]
  }
],
"resourceSpecCharacteristic": [
  {
```


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```
        "name": "virtualization",
        "description": "Type of virtualization required",
        "valueType": "string",
        "resourceSpecCharacteristicValue": [
          {
            "isDefault": true,
            "value": "container-runtime"
          },
          {
            "isDefault": false,
            "value": "hypervisor"
          }
        ]
      }
  ]
}
```


The following is an example of a request to get a SoftwareSupportPackageSpecification:

Request

```
GET /tmf-api/softwareCompute/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7
Accept: application/json
```

Response


```
200

{
  "id": "632547af-e850-45a1-a569-31531ea81eb7",
  "href": "/resourceCatalog/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7",
  "name": "Software Package Repository Specification",
  "description": "This specification describes Software Package in a repository",
  "@type": "SoftwareSupportPackageSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "SoftwareSupportPackage"
  },
  "resourceSpecRelationship": [],
  "resourceSpecCharacteristic": [
    {
      "name": "repoName",
      "description": "Name of the image repository",
      "valueType": "string"
```


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```
    },
    {
      "name": "repoBase",
      "description": "Base URL of the image repository",
      "valueType": "string"
    },
    {
      "name": "imagePath",
      "description": "Path of the image in the repository",
      "valueType": "string"
    }
  ]
}
```

Create resource specification

POST /resourceSpecification

Description

This operation creates a resource specification entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non-mandatory attributes when creating a ResourceSpecification, including any possible rule conditions and applicable default values. Notice that it is up to the implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
name	
@type	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
attachment	
category	
description	
featureSpecification	
isBundle	
lastUpdate	
lifecycleStatus	
relatedParty	
resourceSpecCharacteristic	
resourceSpecRelationship	
targetResourceSchema	
validFor	
version	

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Usage Samples

The following is an example of a request to create a SoftwareSpecification:

Request

```
POST /tmf-api/softwareCompute/v4/resourceSpecification
Content-Type: application/json

{
  "name": "Encabulator Software Specification",
  "description": "This specification describes the Encabulator software",
  "@type": "SoftwareSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "category": "Software",
  "targetResourceSchema": {
    "@type": "InstalledSoftware"
  },
  "relatedParty": [
    {
      "id": "ab5660aa-064d-440c-b7d1-b943e42b904e",
      "href": "/partyManagement/v4/organization/ab5660aa-064d-440c-b7d1-b943e42b904e",
      "role": "supplier",
      "name": "General Electric"
    }
  ],
  "releaseStatus": "firstCustomerShipment",
  "buildNumber": "encabulator-1.0.0",
  "majorVersion": "1",
  "minorVersion": "0",
  "maintenanceVersion": "0",
  "otherDesignator": "alpha",
  "isDistributable": true,
  "installSize": {
    "amount": 9143256,
    "units": "bytes"
  },
  "resourceSpecRelationship": [
    {
      "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "name": "Encabulator Platform Requirement Specification",
      "relationshipType": "requires"
    },
    {
      "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "name": "Encabulator Function Specification",
      "relationshipType": "provides"
    }
  ],
```

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```
  "softwareSupportPackage": {
    "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
    "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
    "name": "Encabulator Software Package",
    "@referredType": "SoftwareSupportPackage"
  },
  "resourceSpecCharacteristic": [
    {
      "name": "OperatingSystem",
      "description": "Operating System",
      "valueType": "String",
      "configurable": true,
      "@type": "ResourceSpecificationCharacteristic",
      "minCardinality": 1,
      "maxCardinality": 3,
      "isUnique": true,
      "extensible": false,
      "resourceSpecCharacteristicValue": [
        {
          "isDefault": true,
          "value": "Debian 11 (bullseye)"
        },
        {
          "isDefault": false,
          "value": "Ubuntu 20.04 LTS (focal)"
        },
        {
          "isDefault": false,
          "value": "RHEL 9"
        }
      ]
    }
  ]
}
```


Response

201


```
{
  "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
  "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
  "name": "Encabulator Software Specification",
  "description": "This specification describes the Encabulator software",
  "@type": "SoftwareSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lastUpdate": "2022-02-03T07:59:59Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Software",
```


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```
  "targetResourceSchema": {
    "@type": "InstalledSoftware"
  },
  "relatedParty": [
    {
      "id": "ab5660aa-064d-440c-b7d1-b943e42b904e",
      "href": "/partyManagement/v4/organization/ab5660aa-064d-440c-b7d1-b943e42b904e",
      "role": "supplier",
      "name": "General Electric"
    }
  ],
  "releaseStatus": "firstCustomerShipment",
  "buildNumber": "encabulator-1.0.0",
  "majorVersion": "1",
  "minorVersion": "0",
  "maintenanceVersion": "0",
  "otherDesignator": "alpha",
  "isDistributable": true,
  "installSize": {
    "amount": 9143256,
    "units": "bytes"
  },
  "isExperimental": false,
  "numUsersMax": 8,
  "numberProcessActiveTotal": 0,
  "resourceSpecRelationship": [
    {
      "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "name": "Encabulator Platform Requirement Specification",
      "relationshipType": "requires"
    },
    {
      "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "name": "Encabulator Function Specification",
      "relationshipType": "provides"
    }
  ],
  "softwareSupportPackage": {
    "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
    "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
    "name": "Encabulator Software Package",
    "@referredType": "SoftwareSupportPackage"
  },
  "resourceSpecCharacteristic": [
    {
      "name": "OperatingSystem",
      "description": "Operating System",
      "valueType": "String",
      "configurable": true,
      "@type": "ResourceSpecificationCharacteristic",
      "minCardinality": 1,
      "maxCardinality": 3,
      "isUnique": true,
```


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```
        "extensible": false,
        "resourceSpecCharacteristicValue": [
          {
            "isDefault": true,
            "value": "Debian 11 (bullseye)"
          },
          {
            "isDefault": false,
            "value": "Ubuntu 20.04 LTS (focal)"
          },
          {
            "isDefault": false,
            "value": "RHEL 9"
          }
        ]
      }
    ]
}
```

The following is an example of a request to create a ResourceFunctionSpecification:

Request

```
POST /tmf-api/softwareCompute/v4/resourceSpecification
Content-Type: application/json

{
  "name": "Encabulator Function Specification",
  "description": "This specification describes the Encabulator Function",
  "@type": "ResourceFunctionSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "ResourceFunction"
  },
  "resourceSpecCharacteristic": [
    {
      "name": "numMarzelvanes",
      "description": "Number of antigravic marzelvanes.",
      "valueType": "integer",
      "resourceSpecCharacteristicValue": [
        {
          "valueFrom": 6,
          "valueTo": 32
        }
      ]
    },
    {
```

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```
        "name": "numStatorSlots",
        "description": "Number of panandermic semi-boloid stator slots.",
        "valueType": "integer",
        "resourceSpecCharacteristicValue": [
          {
            "valueFrom": 0,
            "valueTo": 50
          }
        ]
      }
    ],
    "connectionPointSpecification": [
      {
        "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
        "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
        "name": "TMF635 Specification",
        "@referredType": "APISpecification"
      }
    ]
}
```


Response

201


```
{
    "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
    "name": "Encabulator Function Specification",
    "description": "This specification describes the Encabulator Function",
    "@type": "ResourceFunctionSpecification",
    "validFor": {
      "startDateTime": "2022-11-28T08:00:00Z",
      "endDateTime": "2023-11-29T00:00:00Z"
    },
    "lastUpdate": "2022-02-03T07:59:59Z",
    "lifecycleStatus": "Active",
    "isBundle": false,
    "category": "Software",
    "targetResourceSchema": {
      "@type": "ResourceFunction"
    },
    "resourceSpecRelationship": [],
    "featureSpecification": [],
    "resourceSpecCharacteristic": [
      {
        "name": "numMarzelvanes",
        "description": "Number of antigravic marzelvanes.",
        "valueType": "integer",
        "resourceSpecCharacteristicValue": [
          {
            "valueFrom": 6,
            "valueTo": 32
```


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```
          }
        ]
      },
      {
        "name": "numStatorSlots",
        "description": "Number of panandermic semi-boloid stator slots.",
        "valueType": "integer",
        "resourceSpecCharacteristicValue": [
          {
            "valueFrom": 0,
            "valueTo": 50
          }
        ]
      }
    ],
    "connectionPointSpecification": [
      {
        "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
        "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
        "name": "TMF635 Specification",
        "@referredType": "APISpecification"
      }
    ],
    "connectivitySpecification": []
}
```


The following is an example of a request to create an APISpecification:

Request


```
POST /tmf-api/softwareCompute/v4/resourceSpecification
Content-Type: application/json

{
   "name": "TMF635 Specification",
   "description": "Open API for Usage Management",
   "@type": "APISpecification",
   "validFor": {
      "startDateTime": "2021-07-01T08:00:00Z"
   },
   "category": "Software",
   "targetResourceSchema": {
      "@type": "API"
   },
   "releaseStatus": "generalDeployment",
   "buildNumber": "20.5.0",
   "majorVersion": "4",
   "minorVersion": "0",
   "maintenanceVersion": "0",
   "isDistributable": true,
   "isExperimental": false,
   "apiProtocolType": "REST",
```


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```
  "externalUrl": "https://example.net/tmf-api/usageManagement/v4",
  "internalUrl": "/tmf-api/usageManagement/v4",
  "externalSchema": "https://tmf-open-api-table-documents.s3.eu-west-
1.amazonaws.com/OpenAPITable/4.0.0/swagger/TMF635-UsageManagement-v4.0.0.swagger.json",
  "internalSchema": "/tmf-api/usageManagement/v4/schema/TMF635-UsageManagement-v4.0.0.swagger.json",
  "authenticationType": "Basic",
  "allowedOperations": [
    "GET",
    "HEAD",
    "POST",
    "PATCH",
    "DELETE"
  ],
  "allowedAPIEntities": [
    "usageSpecification",
    "usage"
  ],
  "responseTypeFormat": [
    "application/json",
    "application/problem+json"
  ],
  "relatedParty": [
    {
      "id": "ab5660aa-064d-440c-b7d1-b943e42b904e",
      "href": "/partyManagement/v4/organization/ab5660aa-064d-440c-b7d1-b943e42b904e",
      "name": "TM Forum",
      "role": "supplier"
    }
  ]
}
```


Response

201


```
{
  "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
  "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
  "name": "TMF635 Specification",
  "description": "Open API for Usage Management",
  "@type": "APISpecification",
  "validFor": {
    "startDateTime": "2021-07-01T08:00:00Z"
  },
  "lifecycleStatus": "Active",
  "lastUpdate": "2022-11-27T07:59:59Z",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "API"
  },
  "releaseStatus": "generalDeployment",
  "buildNumber": "20.5.0",
```


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```
  "majorVersion": "4",
  "minorVersion": "0",
  "maintenanceVersion": "0",
  "isDistributable": true,
  "isExperimental": false,
  "apiProtocolType": "REST",
  "externalUrl": "https://example.net/tmf-api/usageManagement/v4",
  "internalUrl": "/tmf-api/usageManagement/v4",
  "externalSchema": "https://tmf-open-api-table-documents.s3.eu-west-
1.amazonaws.com/OpenAPITable/4.0.0/swagger/TMF635-UsageManagement-v4.0.0.swagger.json",
  "internalSchema": "/tmf-api/usageManagement/v4/schema/TMF635-UsageManagement-v4.0.0.swagger.json",
  "authenticationType": "Basic",
  "allowedOperations": [
    "GET",
    "HEAD",
    "POST",
    "PATCH",
    "DELETE"
  ],
  "allowedAPIEntities": [
    "usageSpecification",
    "usage"
  ],
  "responseTypeFormat": [
    "application/json",
    "application/problem+json"
  ],
  "resourceSpecRelationship": [],
  "relatedParty": [
    {
      "id": "ab5660aa-064d-440c-b7d1-b943e42b904e",
      "href": "/partyManagement/v4/organization/ab5660aa-064d-440c-b7d1-b943e42b904e",
      "name": "TM Forum",
      "role": "supplier"
    }
  ],
  "resourceSpecCharacteristic": []
}
```


The following is an example of a request to create a HostingPlatformRequirementSpecification:

Request


```
POST /tmf-api/softwareCompute/v4/resourceSpecification
Content-Type: application/json

{
  "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
  "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
  "name": "Encabulator Platform Requirement Specification",
  "description": "This specification describes requirements on platforms hosting the Encabulator software",
  "@type": "HostingPlatformRequirementSpecification",
```


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```
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "category": "Compute",
  "targetResourceSchema": {
    "@type": "HostingPlatformRequirement"
  },
  "featureSpecification": [
    {
      "name": "scale",
      "featureSpecCharacteristic": [
        {
          "name": "cpuSize",
          "description": "Vertical CPU scaling dimension",
          "valueType": "string",
          "featureSpecCharacteristicValue": [
            {
              "isDefault": true,
              "value": "micro"
            },
            {
              "isDefault": false,
              "value": "small"
            },
            {
              "isDefault": false,
              "value": "medium"
            },
            {
              "isDefault": false,
              "value": "large"
            }
          ]
        },
        {
          "name": "cpuNum",
          "description": "Minimum number of CPUs",
          "valueType": "integer",
          "featureSpecCharacteristicValue": [
            {
              "valueFrom": 1,
              "valueTo": 24
            }
          ]
        },
        {
          "name": "memory",
          "description": "Minimum amount of memory required",
          "valueType": "integer",
          "featureSpecCharacteristicValue": [
            {
              "unitOfMeasure": "MiB",
              "valueFrom": 640,
              "valueTo": 16384
```


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```
              }
            ]
          }
        ]
      }
    ],
    "resourceSpecCharacteristic": [
      {
        "name": "virtualization",
        "description": "Type of virtualization required",
        "valueType": "string",
        "resourceSpecCharacteristicValue": [
          {
            "isDefault": true,
            "value": "container-runtime"
          },
          {
            "isDefault": false,
            "value": "hypervisor"
          }
        ]
      }
    ]
}
```


Response

201


```
{
  "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
  "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
  "name": "Encabulator Platform Requirement Specification",
  "description": "This specification describes requirements on platforms hosting the Encabulator software",
  "@type": "HostingPlatformRequirementSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Compute",
  "targetResourceSchema": {
    "@type": "HostingPlatformRequirement"
  },
  "resourceSpecRelationship": [],
  "featureSpecification": [
    {
      "name": "scale",
      "featureSpecCharacteristic": [
        {
          "name": "cpuSize",
          "description": "Vertical CPU scaling dimension",
```


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```
            "valueType": "string",
            "featureSpecCharacteristicValue": [
              {
                "isDefault": true,
                "value": "micro"
              },
              {
                "isDefault": false,
                "value": "small"
              },
              {
                "isDefault": false,
                "value": "medium"
              },
              {
                "isDefault": false,
                "value": "large"
              }
            ]
          },
          {
            "name": "cpuNum",
            "description": "Minimum number of CPUs",
            "valueType": "integer",
            "featureSpecCharacteristicValue": [
              {
                "valueFrom": 1,
                "valueTo": 24
              }
            ]
          },
          {
            "name": "memory",
            "description": "Minimum amount of memory required",
            "valueType": "integer",
            "featureSpecCharacteristicValue": [
              {
                "unitOfMeasure": "MiB",
                "valueFrom": 640,
                "valueTo": 16384
              }
            ]
          }
        ]
      }
    ],
    "resourceSpecCharacteristic": [
      {
        "name": "virtualization",
        "description": "Type of virtualization required",
        "valueType": "string",
        "resourceSpecCharacteristicValue": [
          {
            "isDefault": true,
            "value": "container-runtime"
```


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```
          },
          {
            "isDefault": false,
            "value": "hypervisor"
          }
        ]
      }
    ]
}
```

The following is an example of a request to create a SoftwareSupportPackageSpecification:

Request

```
POST /tmf-api/softwareCompute/v4/resourceSpecification
Content-Type: application/json

{
   "name": "Software Package Repository Specification",
   "description": "This specification describes Software Package in a repository",
   "@type": "SoftwareSupportPackageSpecification",
   "validFor": {
      "startDateTime": "2022-11-28T08:00:00Z",
      "endDateTime": "2023-11-29T00:00:00Z"
   },
   "category": "Software",
   "targetResourceSchema": {
      "@type": "SoftwareSupportPackage"
   },
   "resourceSpecCharacteristic": [
      {
         "name": "repoName",
         "description": "Name of the image repository",
         "valueType": "string"
      },
      {
         "name": "repoBase",
         "description": "Base URL of the image repository",
         "valueType": "string"
      },
      {
         "name": "imagePath",
         "description": "Path of the image in the repository",
         "valueType": "string"
      }
   ]
}
```

Response

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```
201

{
  "id": "632547af-e850-45a1-a569-31531ea81eb7",
  "href": "/resourceCatalog/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7",
  "name": "Software Package Repository Specification",
  "description": "This specification describes Software Package in a repository",
  "@type": "SoftwareSupportPackageSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "SoftwareSupportPackage"
  },
  "resourceSpecRelationship": [],
  "resourceSpecCharacteristic": [
    {
      "name": "repoName",
      "description": "Name of the image repository",
      "valueType": "string"
    },
    {
      "name": "repoBase",
      "description": "Base URL of the image repository",
      "valueType": "string"
    },
    {
      "name": "imagePath",
      "description": "Path of the image in the repository",
      "valueType": "string"
    }
  ]
}
```

Patch resource specification

`PATCH /resourceSpecification/{id}`

Description

This operation allows partial updates of a resource specification entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

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Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non-patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
@baseType	
@schemaLocation	
attachment	
category	
description	
featureSpecification	
isBundle	
lastUpdate	
lifecycleStatus	
name	
relatedParty	
resourceSpecCharacteristic	
resourceSpecRelationship	
targetResourceSchema	
validFor	
version	

Non Patchable Attributes	Rule
id	
href	
@type	

Usage Samples

Example of updating a SoftwareSpecification using json-patch:

Request

```
PATCH /tmf-api/softwareCompute/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4
Content-Type: application/json-patch+json

[
  {
    "op": "add",
    "path": "resourceSpecCharacteristic/0/resourceSpecCharacteristicValue/2",
    "value": {
      "isDefault": false,
      "value": "Ubuntu 22.04 LTS (jammy)"
    }
  }
]
```

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Response

200


```
{
  "id": "1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
  "href": "/resourceCatalog/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4",
  "name": "Encabulator Software Specification",
  "description": "This specification describes the Encabulator software",
  "@type": "SoftwareSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lastUpdate": "2022-11-27T22:55:15.113Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "InstalledSoftware"
  },
  "relatedParty": [
    {
      "id": "ab5660aa-064d-440c-b7d1-b943e42b904e",
      "href": "/partyManagement/v4/organization/ab5660aa-064d-440c-b7d1-b943e42b904e",
      "role": "supplier",
      "name": "General Electric"
    }
  ],
  "releaseStatus": "firstCustomerShipment",
  "buildNumber": "encabulator-1.0.0",
  "majorVersion": "1",
  "minorVersion": "0",
  "maintenanceVersion": "0",
  "otherDesignator": "alpha",
  "isDistributable": true,
  "installSize": {
    "amount": 9143256,
    "units": "bytes"
  },
  "isExperimental": false,
  "numUsersMax": 8,
  "numberProcessActiveTotal": 0,
  "resourceSpecRelationship": [
    {
      "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
      "name": "Encabulator Platform Requirement Specification",
      "relationshipType": "requires"
    },
    {
      "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
      "name": "Encabulator Function Specification",
```


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```
        "relationshipType": "provides"
      }
    ],
    "softwareSupportPackage": {
      "id": "d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
      "href": "/resourceInventory/v4/resource/d4843e6e-bf6f-409a-85c7-e74afb7a64d7",
      "name": "Encabulator Software Package",
      "@referredType": "SoftwareSupportPackage"
    },
    "resourceSpecCharacteristic": [
      {
        "name": "OperatingSystem",
        "description": "Operating System",
        "valueType": "String",
        "configurable": true,
        "@type": "ResourceSpecificationCharacteristic",
        "minCardinality": 1,
        "maxCardinality": 3,
        "isUnique": true,
        "extensible": false,
        "resourceSpecCharacteristicValue": [
          {
            "isDefault": true,
            "value": "Debian 11 (bullseye)"
          },
          {
            "isDefault": false,
            "value": "Ubuntu 20.04 LTS (focal)"
          },
          {
            "isDefault": false,
            "value": "Ubuntu 22.04 LTS (jammy)"
          },
          {
            "isDefault": false,
            "value": "RHEL 9"
          }
        ]
      }
    ]
}
```


Example of updating a ResourceFunctionSpecification using json-patch:

Request

```
PATCH /tmf-api/softwareCompute/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241
Content-Type: application/json-patch+json

[
  {
    "op": "add",
```

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```
    "path": "/featureSpecification/-",
    "value": {
      "name": "realization",
      "featureSpecCharacteristic": [
        {
          "name": "percentRealization",
          "description": "Target percent realization",
          "valueType": "integer",
          "featureSpecCharacteristicValue": [
            {
              "unitOfMeasure": "%",
              "valueFrom": 0,
              "valueTo": 100
            }
          ]
        },
        {
          "name": "skorMotion",
          "description": "Sinusoidal depleneration",
          "valueType": "string",
          "featureSpecCharacteristicValue": [
            {
              "isDefault": false,
              "value": "low"
            },
            {
              "isDefault": true,
              "value": "medium"
            },
            {
              "isDefault": false,
              "value": "high"
            }
          ]
        }
      ]
    }
  }
]
```


Response

200


```
{
  "id": "68337aed-d83f-4aef-8a7e-064b1fc3a241",
  "href": "/resourceCatalog/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241",
  "name": "Encabulator Function Specification",
  "description": "This specification describes the Encabulator Function",
  "@type": "ResourceFunctionSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
```


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```
  },
  "lastUpdate": "2022-11-27T23:07:55.401Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "ResourceFunction"
  },
  "resourceSpecRelationship": [],
  "featureSpecification": [
    {
      "name": "realization",
      "featureSpecCharacteristic": [
        {
          "name": "percentRealization",
          "description": "Target percent realization",
          "valueType": "integer",
          "featureSpecCharacteristicValue": [
            {
              "unitOfMeasure": "%",
              "valueFrom": 0,
              "valueTo": 100
            }
          ]
        },
        {
          "name": "skorMotion",
          "description": "Sinusoidal depleneration",
          "valueType": "string",
          "featureSpecCharacteristicValue": [
            {
              "isDefault": false,
              "value": "low"
            },
            {
              "isDefault": true,
              "value": "medium"
            },
            {
              "isDefault": false,
              "value": "high"
            }
          ]
        }
      ]
    }
  ],
  "resourceSpecCharacteristic": [
    {
      "name": "numMarzelvanes",
      "description": "Number of antigravic marzelvanes.",
      "valueType": "integer",
      "resourceSpecCharacteristicValue": [
        {
          "valueFrom": 6,
```


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```
            "valueTo": 32
          }
        ]
      },
      {
        "name": "numStatorSlots",
        "description": "Number of panandermic semi-boloid stator slots.",
        "valueType": "integer",
        "resourceSpecCharacteristicValue": [
          {
            "valueFrom": 0,
            "valueTo": 50
          }
        ]
      }
    ],
    "connectionPointSpecification": [
      {
        "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
        "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
        "name": "TMF635 Specification",
        "@referredType": "APISpecification"
      }
    ],
    "connectivitySpecification": []
}
```


Example of updating an APISpecification using json-patch:

Request

```
PATCH /tmf-api/softwareCompute/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/validFor/endDateTime",
    "value": "2023-06-30T23:59:59.999Z"
  }
]
```

Response


```
200

{
  "id": "30dcb35e-b643-4381-a619-df424a3a52c8",
  "href": "/resourceCatalog/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8",
  "name": "TMF635 Specification",
```


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```
  "description": "Open API for Usage Management",
  "@type": "APISpecification",
  "validFor": {
    "startDateTime": "2021-07-01T08:00:00Z",
    "endDateTime": "2023-06-30T23:59:59.999Z"
  },
  "lifecycleStatus": "Active",
  "lastUpdate": "2022-11-27T23:41:37.666Z",
  "isBundle": false,
  "category": "Software",
  "targetResourceSchema": {
    "@type": "API"
  },
  "releaseStatus": "generalDeployment",
  "buildNumber": "20.5.0",
  "majorVersion": "4",
  "minorVersion": "0",
  "maintenanceVersion": "0",
  "isDistributable": true,
  "isExperimental": false,
  "apiProtocolType": "REST",
  "externalUrl": "https://example.net/tmf-api/usageManagement/v4",
  "internalUrl": "/tmf-api/usageManagement/v4",
  "externalSchema": "https://tmf-open-api-table-documents.s3.eu-west-
1.amazonaws.com/OpenAPITable/4.0.0/swagger/TMF635-UsageManagement-v4.0.0.swagger.json",
  "internalSchema": "/tmf-api/usageManagement/v4/schema/TMF635-UsageManagement-v4.0.0.swagger.json",
  "authenticationType": "Basic",
  "allowedOperations": [
    "GET",
    "HEAD",
    "POST",
    "PATCH",
    "DELETE"
  ],
  "allowedAPIEntities": [
    "usageSpecification",
    "usage"
  ],
  "responseTypeFormat": [
    "application/json",
    "application/problem+json"
  ],
  "resourceSpecRelationship": [],
  "relatedParty": [
    {
      "id": "ab5660aa-064d-440c-b7d1-b943e42b904e",
      "href": "/partyManagement/v4/organization/ab5660aa-064d-440c-b7d1-b943e42b904e",
      "name": "TM Forum",
      "role": "supplier"
    }
  ],
  "resourceSpecCharacteristic": []
}
```


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Example of updating an HostingPlatformRequirementSpecification using json-patch:

Request

```
PATCH /tmf-api/softwareCompute/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389
Content-Type: application/json-patch+json

[
  {
    "op": "add",
    "path": "/featureSpecification/0/featureSpecCharacteristicValue/-",
    "value": {
      "isDefault": false,
      "value": "xlarge"
    }
  },
  {
    "op": "replace",
    "path": "/featureSpecification/1/featureSpecCharacteristicValue/0/valueTo",
    "value": 32
  },
  {
    "op": "replace",
    "path": "/featureSpecification/2/featureSpecCharacteristicValue/0/valueTo",
    "value": 32768
  }
]
```

Response

```
200

{
  "id": "de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
  "href": "/resourceCatalog/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389",
  "name": "Encabulator Platform Requirement Specification",
  "description": "This specification describes requirements on platforms hosting the Encabulator software",
  "@type": "HostingPlatformRequirementSpecification",
  "validFor": {
    "startDateTime": "2022-11-28T08:00:00Z",
    "endDateTime": "2023-11-29T00:00:00Z"
  },
  "lifecycleStatus": "Active",
  "isBundle": false,
  "category": "Compute",
  "targetResourceSchema": {
    "@type": "HostingPlatformRequirement"
  },
  "resourceSpecRelationship": [],
  "featureSpecification": [
```

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```
{
  "name": "scale",
  "featureSpecCharacteristic": [
    {
      "name": "cpuSize",
      "description": "Vertical CPU scaling dimension",
      "valueType": "string",
      "featureSpecCharacteristicValue": [
        {
          "isDefault": true,
          "value": "micro"
        },
        {
          "isDefault": false,
          "value": "small"
        },
        {
          "isDefault": false,
          "value": "medium"
        },
        {
          "isDefault": false,
          "value": "large"
        },
        {
          "isDefault": false,
          "value": "xlarge"
        }
      ]
    },
    {
      "name": "cpuNum",
      "description": "Minimum number of CPUs",
      "valueType": "integer",
      "featureSpecCharacteristicValue": [
        {
          "valueFrom": 1,
          "valueTo": 32
        }
      ]
    },
    {
      "name": "memory",
      "description": "Minimum amount of memory required",
      "valueType": "integer",
      "featureSpecCharacteristicValue": [
        {
          "unitOfMeasure": "MiB",
          "valueFrom": 640,
          "valueTo": 32768
        }
      ]
    }
  ]
}
```


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```
  ],
  "resourceSpecCharacteristic": [
    {
      "name": "virtualization",
      "description": "Type of virtualization required",
      "valueType": "string",
      "resourceSpecCharacteristicValue": [
        {
          "isDefault": true,
          "value": "container-runtime"
        },
        {
          "isDefault": false,
          "value": "hypervisor"
        }
      ]
    }
  ]
}
```


Delete resource specification

`DELETE /resourceSpecification/{id}`

Description

This operation deletes a resource specification entity.

Usage Samples

Example of a request for deleting an SoftwareSpecification:

Request
DELETE /tmf-api/softwareCompute/v4/resourceSpecification/1b7fec0f-168c-4513-9ec7-d0233e32b6e4
Response
204

Example of a request for deleting a ResourceFunctionSpecification:

Request

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DELETE /tmf-api/softwareCompute/v4/resourceSpecification/68337aed-d83f-4aef-8a7e-064b1fc3a241
Response
204

Example of a request for deleting an APISpecification:

Request
DELETE /tmf-api/softwareCompute/v4/resourceSpecification/30dcb35e-b643-4381-a619-df424a3a52c8
Response
204

Example of a request for deleting a HostingPlatformRequirementSpecification:

Request
DELETE /tmf-api/softwareCompute/v4/resourceSpecification/de29d1fd-cbf8-4bd8-9e75-3f3905db4389
Response
204

Example of a request for deleting a SoftwareSupportPackageSpecification:

Request
DELETE /tmf-api/softwareCompute/v4/resourceSpecification/632547af-e850-45a1-a569-31531ea81eb7
Response

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204

Operations on Installed Software

Operations on Software Specification

Operations on Resource Function

Operations on Resource Function Specification

Operations on A P I

Operations on A P I Specification

Operations on Hosting Platform Requirement

Operations on Hosting Platform Requirement Specification

Operations on Software Support Package

Operations on Software Support Package Specification

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request
POST /api/hub Accept: application/json {"callback": "http://in.listener.com"}
Response
201 Content-Type: application/json Location: /api/hub/42 {"id":"42","callback":"http://in.listener.com","query":null}

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Unregister listener

`DELETE /hub/{id}`

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request
DELETE /api/hub/42 Accept: application/json
Response
204

Publish Event to listener

`POST /client/listener`

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example “EVENT TYPE” should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

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Request

POST /client/listener
Accept: application/json
 	
{
"event": {
EVENT BODY
},
"eventType": "EVENT_TYPE"
}

Response
201

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Administrative Appendix

Document History

Version History

Version Number	Date	Release led by:	Description
4.0.0	15-Dec-2023	Vance Shipley	part of first release for TMF730 Software and Compute Entity Management API Suite v4.0.0

Release Status

Release Status	Date	Modified by	Description
Production	22-Dec-2023	Pierre Gauthier, TM Forum	Initial release.

Acknowledgements

This document was prepared by members of the Open API team.

Team Member	Company	Role
Vance Shipley	Sigscale	Author

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